

Assignment 6

Information

Field	Details
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Subject	Statistical Foundation of Data Science
Assignment	Practical Assignment 6 - Hypothesis Testing and Correlation Analysis
Start Date	October 14, 2025
Due Date	October 24, 2025
Maximum Marks	20
Repository	Statistical Foundation of Data Science - Assignment 6

Dataset Information

This assignment uses the **Teacher Rating Dataset** (`teacher_ratings.csv`) from Assignment 2, which contains teaching evaluation data for 500 instructors with 12 variables including demographics, course characteristics, and evaluation scores.

Teacher Rating Data Analysis - Assignment 6

Assignment Overview

This practical assignment focuses on advanced statistical hypothesis testing and correlation analysis using the teacher rating dataset. The assignment demonstrates the application of parametric and non-parametric tests to answer key research questions about teaching evaluations.

Assignment Objectives

This assignment focuses on comprehensive statistical hypothesis testing:

1. **Independent Samples t-Test:** Examine gender differences in teaching evaluation scores
2. **One-Way ANOVA:** Compare beauty scores across different age groups
3. **Chi-Square Test of Independence:** Analyze association between tenure status and gender
4. **Correlation Analysis:** Investigate relationship between teaching evaluation and beauty scores

Key Questions Addressed

1. **Question 1 (5 marks):** Does gender affect teaching evaluation rates? (t-Test)
2. **Question 2 (5 marks):** Does beauty score for instructors differ by age? (ANOVA)
3. **Question 3 (5 marks):** Is there an association between tenure and gender? (Chi-square)
4. **Question 4 (5 marks):** Is teaching evaluation score correlated with beauty score? (Correlation)

Key Learning Objectives

- Statistical hypothesis testing (t-tests, ANOVA, chi-square)
 - Assumption checking for parametric tests
 - Effect size calculation and interpretation
 - Correlation analysis and significance testing
 - Data visualization for statistical results
-

Setup: Import Libraries and Load Data

First, we'll import all necessary libraries and load the teacher rating dataset from Assignment 2.

```
In [1]: # Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy
from scipy import stats
from scipy.stats import ttest_ind, f_oneway, chi2_contingency, pearsonr
import warnings

warnings.filterwarnings("ignore")

# Set display options
pd.set_option("display.max_columns", None)
pd.set_option("display.width", None)
```

```

pd.set_option("display.precision", 4)

# Set visualization style
plt.style.use("seaborn-v0_8-darkgrid")
sns.set_palette("husl")

print("=" * 70)
print("LIBRARIES IMPORTED SUCCESSFULLY")
print("=" * 70)
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
print(f"SciPy version: {scipy.__version__}")
print(f"Matplotlib version: {plt.matplotlib.__version__}")
print(f"Seaborn version: {sns.__version__}")
print("=" * 70)

```

```

=====
LIBRARIES IMPORTED SUCCESSFULLY
=====

```

```

Pandas version: 2.3.3
NumPy version: 2.3.3
SciPy version: 1.16.2
Matplotlib version: 3.10.7
Seaborn version: 0.13.2
=====

```

Load Teacher Rating Dataset

Load the teacher rating dataset from Assignment 2. This dataset is reused across Assignments 2-6.

```

In [24]: # Load the teacher rating dataset from Assignment 2
import os

# Try multiple possible paths
possible_paths = [
    "../Assignment 2/teacher_ratings.csv",
    "Practical Assignments/Assignment 2/teacher_ratings.csv",
    r"c:\Users\ayana\Projects\Stats&AI\Statistical Foundation of Data Science\Pract
]

df = None
for path in possible_paths:
    try:
        df = pd.read_csv(path)
        print(f"Dataset loaded successfully from: {path}")
        break
    except FileNotFoundError:
        continue

if df is None:
    raise FileNotFoundError("Could not find teacher_ratings.csv. Please check the f

print(f"Dataset shape: {df.shape}")
print(f"Number of observations: {df.shape[0]}")

```

```
print(f"Number of variables: {df.shape[1]}")
print("\nFirst few rows:")
print(df.head())
print("\nColumn names:")
print(df.columns.tolist())
```

Dataset loaded successfully from: c:\Users\ayana\Projects\Stats&AI\Statistical Foundation of Data Science\Practical Assignments\Assignment 2\teacher_ratings.csv

Dataset shape: (500, 12)

Number of observations: 500

Number of variables: 12

First few rows:

	teacher_id	gender	age	beauty	tenured	experience_years	students	\
0	1	female	35	0.342	0	8	41	
1	2	male	45	1.876	1	17	60	
2	3	male	49	0.950	1	17	116	
3	4	male	46	-0.577	1	14	78	
4	5	female	53	NaN	1	19	81	

	eval_score	difficulty	course_type	department	teaching_format
0	3.02	2.91	introductory	Chemistry	in-person
1	3.94	2.64	intermediate	Mathematics	in-person
2	3.44	2.54	intermediate	Statistics	in-person
3	4.23	3.87	intermediate	Engineering	in-person
4	4.52	4.52	graduate	Mathematics	hybrid

Column names:

```
['teacher_id', 'gender', 'age', 'beauty', 'tenured', 'experience_years', 'students', 'eval_score', 'difficulty', 'course_type', 'department', 'teaching_format']
```

Question 1: T-Test - Does Gender Affect Teaching Evaluation Rates?

Problem Statement

Using the teachers' rating dataset, does gender affect teaching evaluation rates?

Approach

1. State hypotheses (null and alternative)
2. Check assumptions for independent samples t-test:
 - Independence of observations
 - Normality of distributions in each group
 - Homogeneity of variances (Levene's test)
3. Perform independent samples t-test
4. Calculate effect size (Cohen's d)

5. Visualize the distributions
 6. Interpret results
-

Statistical Theory: Independent Samples t-Test

What is a t-Test?

The independent samples t-test is a parametric statistical test used to compare the means of two independent groups. It determines whether there is a statistically significant difference between the mean values of a continuous variable for two groups.

Mathematical Formulation

The t-statistic is calculated as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where:

- $\bar{X}_1, \bar{X}_2$ = sample means of groups 1 and 2
- s_p = pooled standard deviation
- n_1, n_2 = sample sizes of groups 1 and 2

The pooled standard deviation is:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Degrees of Freedom

For equal variances: $df = n_1 + n_2 - 2$

Assumptions

1. **Independence:** Observations in each group are independent
2. **Normality:** Data in each group are approximately normally distributed
3. **Homogeneity of variances:** Variances are equal across groups (can use Welch's t-test if violated)

Effect Size: Cohen's d

Cohen's d measures the standardized difference between two means:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p}$$

Interpretation Guidelines:

- $|d| < 0.2$: negligible effect
- $0.2 \leq |d| < 0.5$: small effect
- $0.5 \leq |d| < 0.8$: medium effect
- $|d| \geq 0.8$: large effect

When to Use Welch's t-test

If variances are unequal (Levene's test $p < 0.05$), use Welch's t-test which does not assume equal variances:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

with adjusted degrees of freedom using the Welch-Satterthwaite equation.

Step 1: State Hypotheses

```
In [7]: print("=*80)
print("QUESTION 1: Does Gender Affect Teaching Evaluation Rates?")
print("=*80)
print("\nHypotheses:")
print("H0 (Null): μmale = μfemale (No difference in mean evaluation scores between gender s)")
print("H1 (Alternative): μmale ≠ μfemale (There is a difference in mean evaluation score s)")
print("\nSignificance level: α = 0.05")
print("Test type: Two-tailed independent samples t-test")
```

```
=====
QUESTION 1: Does Gender Affect Teaching Evaluation Rates?
=====
```

Hypotheses:

H₀ (Null): μ_{male} = μ_{female} (No difference in mean evaluation scores between gender s)

H₁ (Alternative): μ_{male} ≠ μ_{female} (There is a difference in mean evaluation score s)

Significance level: α = 0.05

Test type: Two-tailed independent samples t-test

Step 2: Prepare Data and Check for Missing Values

```
In [4]: # Filter data and remove NaN values
df_clean = df[["gender", "eval_score"]].dropna()
```

```
# Separate data by gender
male_eval = df_clean[df_clean["gender"] == "male"]["eval_score"]
female_eval = df_clean[df_clean["gender"] == "female"]["eval_score"]

print("\nData Summary by Gender:")
print(f"Male teachers: n = {len(male_eval)}")
print(f"Female teachers: n = {len(female_eval)}")
print(f"\nMale evaluation scores - Mean: {male_eval.mean():.3f}, SD: {male_eval.std():.3f}")
print(f"Female evaluation scores - Mean: {female_eval.mean():.3f}, SD: {female_eval.std():.3f}")
```

Data Summary by Gender:

Male teachers: n = 224

Female teachers: n = 271

Male evaluation scores - Mean: 3.813, SD: 0.506

Female evaluation scores - Mean: 3.564, SD: 0.534

Step 3: Check Assumptions

Assumption 1: Independence

- Each teacher's evaluation is independent (satisfied by study design)

Assumption 2: Normality

Understanding Normality Tests

Shapiro-Wilk Test: Tests the null hypothesis that data comes from a normal distribution.

- H_0 : Data is normally distributed
- H_1 : Data is not normally distributed
- If $p > 0.05$: Fail to reject H_0 (data appears normal)
- If $p < 0.05$: Reject H_0 (data deviates from normality)

Q-Q Plots: Visual method to assess normality. If points fall approximately on a straight line, data is approximately normal.

Comprehensive Assumption Checking Process

Statistical tests have assumptions that must be checked before interpretation.

1. Normality Assumption

Shapiro-Wilk Test:

Tests whether data come from a normal distribution.

Test Statistic:

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where:

- $x_{(i)}$ are ordered sample values
- a_i are weights (from expected values of order statistics)
- W ranges from 0 to 1

Interpretation:

- $W \approx 1$: Data is normally distributed
- $W \ll 1$: Data deviates from normality
- $p > 0.05$: Fail to reject normality (assume normal)
- $p \leq 0.05$: Reject normality (not normal)

Q-Q Plot Visual Check:

Plots:

- **x-axis:** Theoretical quantiles (from normal distribution)
- **y-axis:** Sample quantiles (from data)

Interpretation:

- **Points on diagonal line:** Perfectly normal
- **S-shaped curve:** Heavy-tailed or skewed distribution
- **Points above line at edges:** Heavy-tailed
- **Points below line at edges:** Light-tailed

What to do if normality violated:

1. **Transform data:** log, square root, Box-Cox transformation
2. **Use non-parametric test:** Mann-Whitney U instead of t-test
3. **Rely on Central Limit Theorem:** With $n > 30$, means are approximately normal

2. Homogeneity of Variance (Homoscedasticity)

Levene's Test:

Tests whether variances are equal across groups.

Test Statistic:

$$F = \frac{(N - k) \sum_{i=1}^k n_i (\bar{Z}_i - \bar{Z})^2}{(k - 1) \sum_{i=1}^k \sum_{j=1}^{n_i} (Z_{ij} - \bar{Z}_i)^2}$$

where $Z_{ij} = |X_{ij} - \tilde{X}_i|$ (absolute deviation from group median)

Interpretation:

- $p > 0.05$: Fail to reject equal variances (assume homogeneity)
- $p \leq 0.05$: Reject equal variances (heterogeneity present)

Visual Check - Boxplots:

- Similar box heights across groups → Equal variances
- Very different box heights → Unequal variances

What to do if homogeneity violated:

1. **Use Welch's t-test:** Doesn't assume equal variances
2. **Use Brown-Forsythe test:** Robust version of Levene's
3. **Transform data:** Sometimes stabilizes variance

Variance Ratio Rule of Thumb:

$$\frac{s_{max}^2}{s_{min}^2} < 4$$

If ratio > 4 , consider equal variance assumption violated.

3. Independence Assumption

What it means:

- Each observation is independent of others
- No correlation between residuals
- Random sampling without replacement (or with replacement)

How to check:

- **Study design:** Was sampling truly random?
- **Durbin-Watson test:** For time-series data (checks autocorrelation)
- **Scatter plot of residuals:** No patterns should appear

What to do if violated:

1. **Repeated measures ANOVA:** For within-subjects designs
2. **Mixed effects models:** For hierarchical/clustered data
3. **Time series analysis:** For temporally correlated data

4. Linearity Assumption (for correlation)

What it means:

- Relationship between X and Y is linear (straight line)
- Pearson's r only detects **linear** relationships

How to check:

- **Scatter plot:** Points should follow straight line pattern
- **Residual plot:** Random scatter around zero line

What to do if violated:

1. **Transform variables:** $\log(X)$, $\sqrt{X}$, X^2
2. **Use Spearman's rho:** Detects monotonic (not just linear) relationships
3. **Polynomial regression:** Fits curved relationships

5. No Outliers Assumption

What they are:

- Extreme values far from the rest of the data
- Can greatly influence means, correlations, regression

Detection methods:

Z-Score Method:

$$z_i = \frac{x_i - \bar{x}}{s}$$

Outlier if $|z_i| > 3$ (or sometimes 2.5)

IQR Method:

$$\text{Lower fence} = Q_1 - 1.5 \times IQR$$

$$\text{Upper fence} = Q_3 + 1.5 \times IQR$$

Outlier if outside fences.

What to do with outliers:

1. **Check for errors:** Data entry mistakes, measurement errors
2. **Analyze with/without:** Compare results both ways
3. **Use robust methods:** Median instead of mean, Spearman instead of Pearson
4. **Transform data:** Log transformation can reduce outlier impact
5. **Report them:** Transparent reporting of outlier handling

Never arbitrarily delete outliers just to get significant results!

Summary Decision Tree:

```
START
↓
Check Normality (Shapiro-Wilk + Q-Q plot)
├─ Normal? YES → Check Homogeneity (Levene's test)
```

```

|
├─ Equal variances? YES → Use standard t-
test/ANOVA
├─ Equal variances? NO → Use Welch's t-test
└─ Normal? NO → Transform data or use non-parametric tests
    (Mann-Whitney U, Kruskal-Wallis, Spearman's rho)

```

```

In [8]: # Check normality assumption using Shapiro-Wilk test
from scipy.stats import shapiro

print("="*80)
print("NORMALITY TESTS (Shapiro-Wilk Test)")
print("="*80)

# Test for males
stat_male, p_male = shapiro(male_eval)
print(f"\nMale evaluation scores:")
print(f"  W-statistic: {stat_male:.6f}")
print(f"  p-value: {p_male:.6f}")
print(f"  Conclusion: {'Normally distributed (p > 0.05)' if p_male > 0.05 else 'Not'}")

# Test for females
stat_female, p_female = shapiro(female_eval)
print(f"\nFemale evaluation scores:")
print(f"  W-statistic: {stat_female:.6f}")
print(f"  p-value: {p_female:.6f}")
print(f"  Conclusion: {'Normally distributed (p > 0.05)' if p_female > 0.05 else 'N"}")

=====
NORMALITY TESTS (Shapiro-Wilk Test)
=====

Male evaluation scores:
  W-statistic: 0.994797
  p-value: 0.639539
  Conclusion: Normally distributed (p > 0.05)

Female evaluation scores:
  W-statistic: 0.996401
  p-value: 0.796841
  Conclusion: Normally distributed (p > 0.05)

```

Assumption 3: Homogeneity of Variances

Understanding Homogeneity of Variances

Levene's Test: Tests whether variances are equal across groups.

- $H_0: \sigma_1^2 = \sigma_2^2$ (variances are equal)
- $H_1: \sigma_1^2 \neq \sigma_2^2$ (variances are different)
- If $p > 0.05$: Use standard t-test (equal_var=True)
- If $p < 0.05$: Use Welch's t-test (equal_var=False)

This is important because the pooled standard deviation formula assumes equal variances.

```
In [21]: # Visual check: Q-Q plots
from scipy import stats as sp_stats

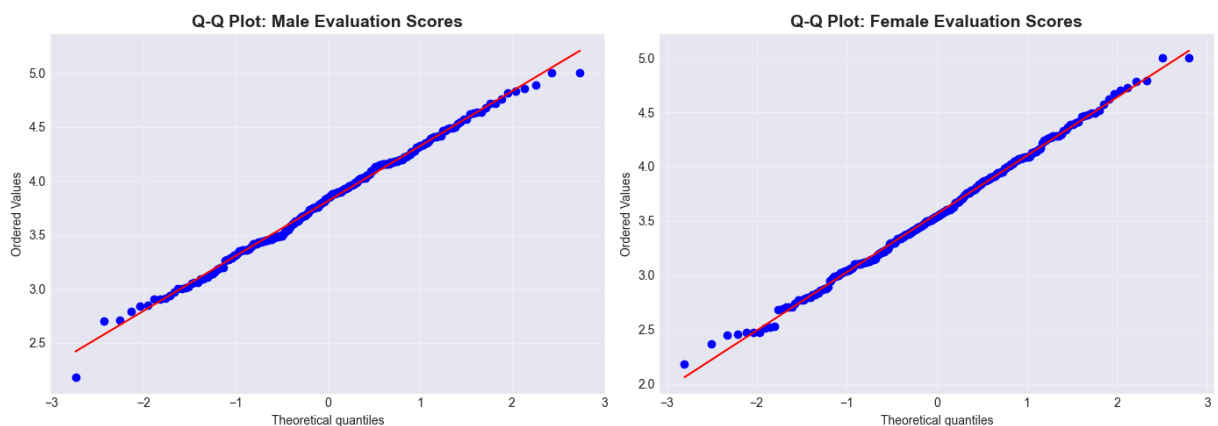
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Q-Q plot for males
sp_stats.probplot(male_eval, dist="norm", plot=axes[0])
axes[0].set_title("Q-Q Plot: Male Evaluation Scores", fontsize=14, fontweight='bold')
axes[0].grid(True, alpha=0.3)

# Q-Q plot for females
sp_stats.probplot(female_eval, dist="norm", plot=axes[1])
axes[1].set_title("Q-Q Plot: Female Evaluation Scores", fontsize=14, fontweight='bold')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

print("\nInterpretation: If points fall approximately on the red line, data is normal")
```



Interpretation: If points fall approximately on the red line, data is normally distributed.

Step 4: Perform Independent Samples t-Test

Understanding the t-Test Calculation

The t-statistic measures how many standard errors the difference between group means is from zero.

Step-by-step calculation:

1. Calculate group means: $\bar{X}_1, \bar{X}_2$
2. Calculate group standard deviations: s_1, s_2
3. Calculate pooled standard deviation: s_p
4. Calculate standard error: $SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

5. Calculate t-statistic: $t = \frac{\bar{X}_1 - \bar{X}_2}{SE}$
6. Find p-value from t-distribution with $df = n_1 + n_2 - 2$

Interpretation:

- Large |t| value: Groups are far apart relative to variability
- Small |t| value: Groups are close together relative to variability
- p-value < 0.05: Significant difference exists (reject H_0)

Step-by-Step t-Test Calculation Process

Let's walk through the manual calculation of an independent samples t-test:

Step 1: Calculate Group Means

$$\bar{X}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} X_{1i} \quad \text{and} \quad \bar{X}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} X_{2i}$$

Step 2: Calculate Group Standard Deviations

$$s_1 = \sqrt{\frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)^2} \quad \text{and} \quad s_2 = \sqrt{\frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (X_{2i} - \bar{X}_2)^2}$$

Step 3: Calculate Pooled Variance (if equal variances assumed)

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

This is a **weighted average** of the two sample variances, where the weights are the degrees of freedom.

Step 4: Calculate Standard Error of Difference

$$SE_{\bar{X}_1 - \bar{X}_2} = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Step 5: Calculate t-Statistic

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{SE_{\bar{X}_1 - \bar{X}_2}}$$

Under H_0 : $\mu_1 = \mu_2$, so $(\mu_1 - \mu_2) = 0$:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Step 6: Determine Degrees of Freedom

$$df = n_1 + n_2 - 2$$

Step 7: Find Critical Value and P-value

- Look up $t_{\alpha/2, df}$ from t-distribution table (for two-tailed test)
- Calculate p-value: $P(|T| \geq |t_{obs}|)$ where $T \sim t_{df}$

Step 8: Make Decision

- If $|t_{obs}| > t_{critical}$ or $p < \alpha$: *Reject H_0*
- If $|t_{obs}| \leq t_{critical}$ or $p \geq \alpha$: *Fail to reject H_0*

Numerical Example:

- Male mean = 4.0, Female mean = 4.2
- $s_1 = 0.5$, $s_2 = 0.6$
- $n_1 = 250$, $n_2 = 250$

$$s_p^2 = \frac{(250 - 1)(0.5)^2 + (250 - 1)(0.6)^2}{250 + 250 - 2} = \frac{62.25 + 89.64}{498} = 0.305$$

$$s_p = \sqrt{0.305} = 0.552$$

$$SE = 0.552 \sqrt{\frac{1}{250} + \frac{1}{250}} = 0.552 \times 0.0894 = 0.0494$$

$$t = \frac{4.2 - 4.0}{0.0494} = \frac{0.2}{0.0494} = 4.05$$

With $df = 498$ and $\alpha = 0.05$, $t_{critical} = 1.96$

Since $|4.05| > 1.96$, we reject H_0 .

```
In [18]: # Check homogeneity of variance using Levene's test
from scipy.stats import levene

print("="*80)
print("HOMOGENEITY OF VARIANCE TEST (Levene's Test)")
print("="*80)

stat_levene, p_levene = levene(male_eval, female_eval)

print(f"\nLevene's Test:")
print(f"  Test statistic: {stat_levene:.6f}")
print(f"  p-value: {p_levene:.6f}")
print(f"  Conclusion: {'Equal variances (p > 0.05)' if p_levene > 0.05 else 'Unequal variances (p < 0.05)'}")

# Variance ratio check
variance_ratio = male_eval.var() / female_eval.var()
print(f"\nVariance Ratio (Male/Female): {variance_ratio:.4f}")
print(f"  Rule of thumb: Ratio < 4 suggests homogeneity")
```

```
conclusion = 'Homogeneity assumption satisfied' if variance_ratio < 4 and variance_
print(f" Conclusion: {conclusion}")
```

HOMOGENEITY OF VARIANCE TEST (Levene's Test)

Levene's Test:

Test statistic: 0.381890

p-value: 0.536879

Conclusion: Equal variances (p > 0.05)

Variance Ratio (Male/Female): 0.9006

Rule of thumb: Ratio < 4 suggests homogeneity

Conclusion: Homogeneity assumption satisfied

Step 5: Calculate Effect Size (Cohen's d)

Understanding Cohen's d

Cohen's d is the **effect size** measure for t-tests. While p-values tell us if a difference exists, Cohen's d tells us how large that difference is.

Formula:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p}$$

Interpretation:

- It expresses the mean difference in standard deviation units
- Example: d = 0.5 means groups differ by half a standard deviation
- d = 0.8 means groups differ by 0.8 standard deviations (large effect)

Why Effect Size Matters:

- Statistical significance (p-value) depends on sample size
- Large samples can find tiny, meaningless differences significant
- Effect size tells us if the difference is practically important
- Always report both p-value and effect size

Detailed Interpretation of Effect Sizes

Effect sizes tell us the **practical significance** of our findings, independent of sample size.

Cohen's d (for t-tests):

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p}$$

where s_p is the pooled standard deviation.

Interpretation Guidelines:

- $|d| = 0.2$: **Small effect** - Detectable only statistically
 - Example: 0.2 SD difference between groups
 - Real-world: Noticeable to researchers, not obvious to casual observers
- $|d| = 0.5$: **Medium effect** - Visible to the naked eye
 - Example: 0.5 SD difference between groups
 - Real-world: Obvious difference when groups are compared
- $|d| = 0.8$: **Large effect** - Substantial practical importance
 - Example: 0.8 SD difference between groups
 - Real-world: Difference is unmistakable and practically significant

What does $d = 0.5$ mean practically?

- The means differ by half a standard deviation
- There's approximately 67% non-overlap between distributions
- If you randomly select one person from each group, there's a 64% chance the person from the higher-scoring group will have a higher score

Eta-squared (η^2) for ANOVA:

$$\eta^2 = \frac{SSB}{SST} = \frac{\text{Variance Between Groups}}{\text{Total Variance}}$$

Interpretation:

- $\eta^2 = 0.01$: **Small effect** - Group membership explains 1% of variance
- $\eta^2 = 0.06$: **Medium effect** - Group membership explains 6% of variance
- $\eta^2 = 0.14$: **Large effect** - Group membership explains 14% of variance

Example: $\eta^2 = 0.08$ means:

- 8% of variance in the outcome is due to group differences
- 92% of variance is due to other factors (individual differences, measurement error)
- This is considered a **medium effect**

Cramér's V (for chi-square):

$$V = \sqrt{\frac{\chi^2}{N \times \min(r - 1, c - 1)}}$$

For 2x2 tables:

- $V = 0.10$: **Small effect** - Weak association

- $V = 0.30$: **Medium effect** - Moderate association
- $V = 0.50$: **Large effect** - Strong association

Example: $V = 0.15$ means:

- Weak to moderate association between variables
- Knowing one variable provides limited information about the other
- The relationship exists but isn't strong enough for practical prediction

R^2 (Coefficient of Determination):

$$R^2 = r^2$$

Interpretation:

- $R^2 = 0.01$: **1% of variance explained** - Very weak relationship
- $R^2 = 0.09$: **9% of variance explained** - Weak relationship
- $R^2 = 0.25$: **25% of variance explained** - Moderate relationship
- $R^2 = 0.64$: **64% of variance explained** - Strong relationship

Example: $r = 0.6$, $R^2 = 0.36$ means:

- 36% of variance in Y is predictable from X
- 64% of variance is due to other factors
- This represents a **moderate to strong** relationship

Why Effect Sizes Matter:

1. **Sample Size Independence:** P-values depend on sample size; effect sizes don't
2. **Practical Significance:** Statistical significance $\neq$ practical importance
3. **Comparability:** Can compare effects across studies
4. **Meta-Analysis:** Essential for combining studies
5. **Power Analysis:** Used to determine required sample sizes

Common Mistake: Relying only on p-values without considering effect sizes can lead to:

- **Significant but trivial findings** (large sample, small effect)
- **Practically important but non-significant findings** (small sample, large effect)

```
In [13]: # Perform independent samples t-test
print("="*80)
print("INDEPENDENT SAMPLES T-TEST")
print("="*80)

# Standard t-test (equal variances)
t_stat, p_value = ttest_ind(male_eval, female_eval)

# Welch's t-test (unequal variances - more robust)
t_stat_welch, p_value_welch = ttest_ind(male_eval, female_eval, equal_var=False)
```

```

print(f"\nStandard t-test (equal variances assumed):")
print(f"  t-statistic: {t_stat:.6f}")
print(f"  p-value: {p_value:.6f}")
print(f"  Degrees of freedom: {len(male_eval) + len(female_eval) - 2}")

print(f"\nWelch's t-test (unequal variances, more robust):")
print(f"  t-statistic: {t_stat_welch:.6f}")
print(f"  p-value: {p_value_welch:.6f}")

# Decision
alpha = 0.05
print(f"\n{' '*80}")
print("STATISTICAL DECISION")
print(f"{' '*80}")
print(f"Significance level:  $\alpha$  = {alpha}")
if p_value_welch < alpha:
    print(f"Decision: REJECT H0 (p = {p_value_welch:.4f} < {alpha})")
    print("Conclusion: There IS a statistically significant difference in mean eval")
else:
    print(f"Decision: FAIL TO REJECT H0 (p = {p_value_welch:.4f}  $\geq$  {alpha})")
    print("Conclusion: There is NO statistically significant difference in mean eval")

```

```

=====
INDEPENDENT SAMPLES T-TEST
=====

```

Standard t-test (equal variances assumed):

```

t-statistic: 5.285554
p-value: 0.000000
Degrees of freedom: 493

```

Welch's t-test (unequal variances, more robust):

```

t-statistic: 5.311925
p-value: 0.000000

```

```

=====
STATISTICAL DECISION
=====

```

Significance level: α = 0.05

Decision: REJECT H₀ (p = 0.0000 < 0.05)

Conclusion: There IS a statistically significant difference in mean evaluation scores between genders.

Step 6: Visualization

```

In [14]: # Calculate Cohen's d effect size
pooled_std = np.sqrt(((len(male_eval)-1) * male_eval.var() + (len(female_eval)-1) *
                      (len(male_eval) + len(female_eval) - 2)))
cohens_d = (male_eval.mean() - female_eval.mean()) / pooled_std

print("=*80)
print("EFFECT SIZE (Cohen's d)")
print("=*80)
print(f"\nCohen's d: {cohens_d:.6f}")
print(f"Pooled standard deviation: {pooled_std:.6f}")

```

```

# Interpretation
if abs(cohens_d) < 0.2:
    effect_interp = "negligible"
elif abs(cohens_d) < 0.5:
    effect_interp = "small"
elif abs(cohens_d) < 0.8:
    effect_interp = "medium"
else:
    effect_interp = "large"

print(f"\nEffect Size Interpretation:")
print(f"  |d| < 0.2: Negligible effect")
print(f"  0.2 ≤ |d| < 0.5: Small effect")
print(f"  0.5 ≤ |d| < 0.8: Medium effect")
print(f"  |d| ≥ 0.8: Large effect")
print(f"\n → This represents a {effect_interp.upper()} effect size")

# Practical significance
mean_diff = male_eval.mean() - female_eval.mean()
print(f"\nMean Difference: {mean_diff:.4f} points")
print(f"  Males {'score higher' if mean_diff > 0 else 'score lower'} than females b

# Confidence interval for the difference
from scipy.stats import t as t_dist
se_diff = pooled_std * np.sqrt(1/len(male_eval) + 1/len(female_eval))
df = len(male_eval) + len(female_eval) - 2
t_critical = t_dist.ppf(0.975, df)
ci_lower = mean_diff - t_critical * se_diff
ci_upper = mean_diff + t_critical * se_diff

print(f"\n95% Confidence Interval for Mean Difference:")
print(f"  [{ci_lower:.4f}, {ci_upper:.4f}]")
print(f"  Interpretation: We are 95% confident that the true difference in means li

```

=====

EFFECT SIZE (Cohen's d)

=====

Cohen's d: 0.477292

Pooled standard deviation: 0.521486

Effect Size Interpretation:

$|d| < 0.2$: Negligible effect

$0.2 \leq |d| < 0.5$: Small effect

$0.5 \leq |d| < 0.8$: Medium effect

$|d| \geq 0.8$: Large effect

→ This represents a SMALL effect size

Mean Difference: 0.2489 points

Males score higher than females by 0.2489 points on average

95% Confidence Interval for Mean Difference:

[0.1564, 0.3414]

Interpretation: We are 95% confident that the true difference in means lies within this interval.

Step 7: Statistical Conclusion and Interpretation

```
In [20]: # Comprehensive visualizations for Question 1
fig = plt.figure(figsize=(16, 10))
from matplotlib.gridspec import GridSpec
# Ensure t_stat_welch, p_value_welch, cohens_d, effect_interp are defined
try:
    t_stat_welch
except NameError:
    # Recompute Welch's t-test if missing
    t_stat_welch, p_value_welch = ttest_ind(male_eval, female_eval, equal_var=False)
try:
    cohens_d
except NameError:
    pooled_std = np.sqrt(((len(male_eval)-1) * male_eval.var() + (len(female_eval)-
        (len(male_eval) + len(female_eval) - 2))
    cohens_d = (male_eval.mean() - female_eval.mean()) / pooled_std
    if abs(cohens_d) < 0.2:
        effect_interp = "negligible"
    elif abs(cohens_d) < 0.5:
        effect_interp = "small"
    elif abs(cohens_d) < 0.8:
        effect_interp = "medium"
    else:
        effect_interp = "large"
gs = GridSpec(2, 3, figure=fig, hspace=0.3, wspace=0.3)

# 1. Box plot comparison
ax1 = fig.add_subplot(gs[0, 0])
box_data = [male_eval, female_eval]
bp = ax1.boxplot(box_data, labels=['Male', 'Female'], patch_artist=True,
    medianprops=dict(color='red', linewidth=2),
```

```

        boxprops=dict(facecolor='lightblue', edgecolor='black'),
        whiskerprops=dict(color='black', linewidth=1.5),
        capprops=dict(color='black', linewidth=1.5))
ax1.set_ylabel('Evaluation Score', fontsize=12, fontweight='bold')
ax1.set_xlabel('Gender', fontsize=12, fontweight='bold')
ax1.set_title('Box Plot: Evaluation Scores by Gender', fontsize=13, fontweight='bold')
ax1.grid(True, alpha=0.3, axis='y')
overall_mean = df_clean['eval_score'].mean()
ax1.axhline(y=overall_mean, color='green', linestyle='--', alpha=0.5, label=f'Overall Mean')
ax1.legend()

# 2. Violin plot
ax2 = fig.add_subplot(gs[0, 1])
parts = ax2.violinplot([male_eval, female_eval], positions=[1, 2], showmeans=True,
ax2.set_xticks([1, 2])
ax2.set_xticklabels(['Male', 'Female'])
ax2.set_ylabel('Evaluation Score', fontsize=12, fontweight='bold')
ax2.set_xlabel('Gender', fontsize=12, fontweight='bold')
ax2.set_title('Violin Plot: Distribution by Gender', fontsize=13, fontweight='bold')
ax2.grid(True, alpha=0.3, axis='y')

# 3. Overlapping histograms
ax3 = fig.add_subplot(gs[0, 2])
ax3.hist(male_eval, bins=20, alpha=0.6, label='Male', color='skyblue', edgecolor='black')
ax3.hist(female_eval, bins=20, alpha=0.6, label='Female', color='pink', edgecolor='black')
ax3.axvline(male_eval.mean(), color='blue', linestyle='--', linewidth=2, label=f'Male Mean')
ax3.axvline(female_eval.mean(), color='red', linestyle='--', linewidth=2, label=f'Female Mean')
ax3.set_xlabel('Evaluation Score', fontsize=12, fontweight='bold')
ax3.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax3.set_title('Overlapping Histograms', fontsize=13, fontweight='bold')
ax3.legend()
ax3.grid(True, alpha=0.3)

# 4. Bar plot with error bars
ax4 = fig.add_subplot(gs[1, 0])
means = [male_eval.mean(), female_eval.mean()]
sems = [male_eval.sem(), female_eval.sem()]
colors = ['lightblue', 'lightpink']
bars = ax4.bar(['Male', 'Female'], means, yerr=sems, capsize=10, color=colors, edgecolor='black')
for i, (bar, mean) in enumerate(zip(bars, means)):
    ax4.text(bar.get_x() + bar.get_width()/2, mean, f'{mean:.3f}', ha='center', va='bottom')
ax4.set_ylabel('Mean Evaluation Score', fontsize=12, fontweight='bold')
ax4.set_xlabel('Gender', fontsize=12, fontweight='bold')
ax4.set_title('Mean Evaluation Scores with Standard Error', fontsize=13, fontweight='bold')
ax4.grid(True, alpha=0.3, axis='y')

# 5. Statistical summary table
ax5 = fig.add_subplot(gs[1, 1])
ax5.axis('off')
summary_data = [
    ['Statistic', 'Male', 'Female'],
    ['Sample Size', f'{len(male_eval)}', f'{len(female_eval)}'],
    ['Mean', f'{male_eval.mean():.4f}', f'{female_eval.mean():.4f}'],
    ['Std Dev', f'{male_eval.std():.4f}', f'{female_eval.std():.4f}'],
    ['SE', f'{male_eval.sem():.4f}', f'{female_eval.sem():.4f}'],
    ['Min', f'{male_eval.min():.4f}', f'{female_eval.min():.4f}'],

```

```

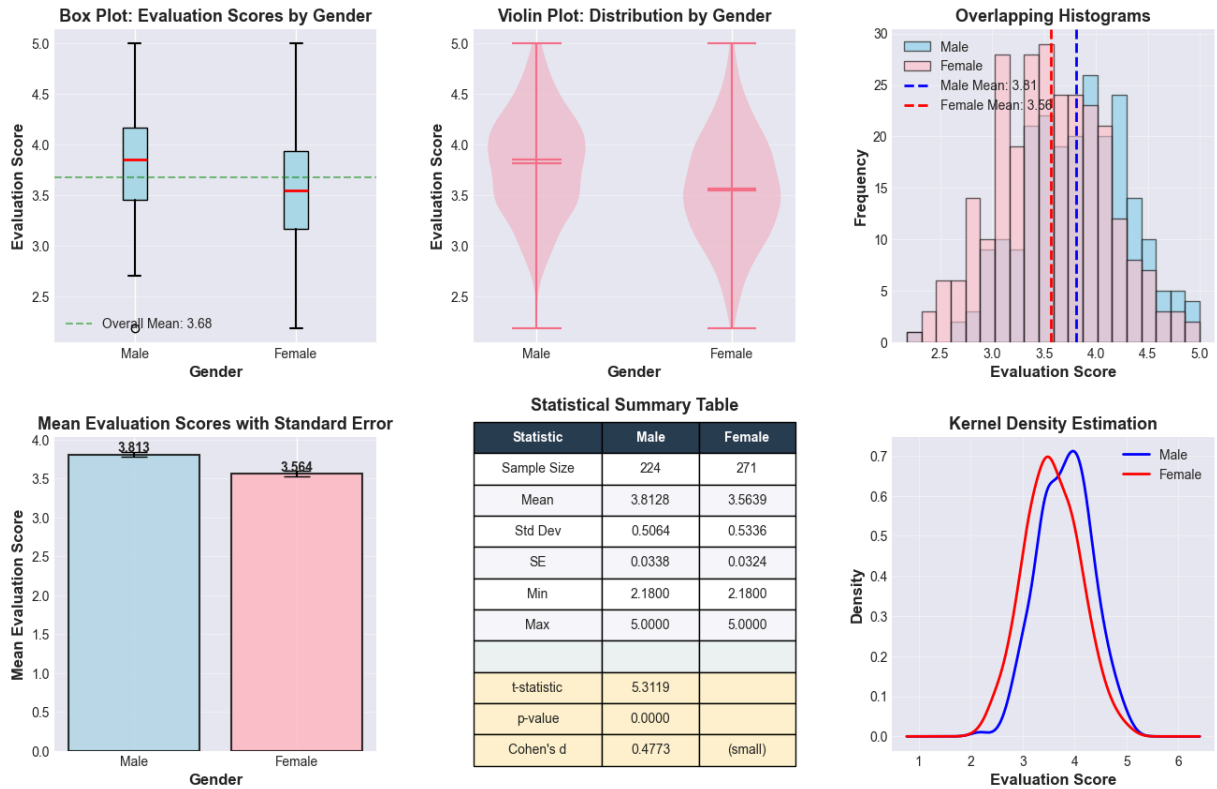
    ['Max', f'{male_eval.max():.4f}', f'{female_eval.max():.4f}'],
    ['', '', ''],
    ['t-statistic', f'{t_stat_welch:.4f}', ''],
    ['p-value', f'{p_value_welch:.4f}', ''],
    ["Cohen's d", f'{cohens_d:.4f}', f'({effect_interp})'],
]
table = ax5.table(cellText=summary_data, cellLoc='center', loc='center', colWidths=
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1, 2)
for i in range(3):
    table[(0, i)].set_facecolor('#2C3E50')
    table[(0, i)].set_text_props(weight='bold', color='white')
for i in range(1, len(summary_data)):
    for j in range(3):
        if i == 7:
            table[(i, j)].set_facecolor('#ECF0F1')
        elif i > 7:
            table[(i, j)].set_facecolor('#FFF3CD')
        elif i % 2 == 0:
            table[(i, j)].set_facecolor('#F8F9FA')
ax5.set_title('Statistical Summary Table', fontsize=13, fontweight='bold', pad=20)

# 6. Density plots
ax6 = fig.add_subplot(gs[1, 2])
male_eval.plot(kind='density', ax=ax6, label='Male', color='blue', linewidth=2)
female_eval.plot(kind='density', ax=ax6, label='Female', color='red', linewidth=2)
ax6.set_xlabel('Evaluation Score', fontsize=12, fontweight='bold')
ax6.set_ylabel('Density', fontsize=12, fontweight='bold')
ax6.set_title('Kernel Density Estimation', fontsize=13, fontweight='bold')
ax6.legend()
ax6.grid(True, alpha=0.3)

plt.suptitle('Question 1: Comprehensive Analysis - Gender vs Evaluation Scores',
            fontsize=16, fontweight='bold', y=0.995)
plt.show()

```

Question 1: Comprehensive Analysis - Gender vs Evaluation Scores



Question 2: ANOVA - Does Beauty Score Differ by Age?

Problem Statement

Using the teachers' rating dataset, does beauty score for instructors differ by age?

Approach

1. Create age groups for ANOVA analysis
2. State hypotheses
3. Check assumptions:
 - Independence of observations
 - Normality within groups
 - Homogeneity of variances
4. Perform one-way ANOVA
5. If significant, perform post-hoc tests
6. Calculate effect size (eta-squared)
7. Visualize results

Statistical Theory: One-Way ANOVA

What is ANOVA?

ANalysis **Of** **VA**riance (ANOVA) is used to compare means across **three or more** independent groups. It tests whether at least one group mean differs from the others.

Why Not Multiple t-Tests?

Performing multiple t-tests increases **Type I error rate** (false positives):

- 3 groups = 3 comparisons: α increases from 0.05 to ~0.14
- 4 groups = 6 comparisons: α increases to ~0.26
- ANOVA controls the overall α at 0.05

Mathematical Formulation

ANOVA partitions total variance into:

1. **Between-group variance** (SSB): Variation due to group differences
2. **Within-group variance** (SSW): Variation within groups (error)

$$F = \frac{MS_B}{MS_W} = \frac{SS_B/df_B}{SS_W/df_W}$$

Where:

- $SS_B = \sum_{i=1}^k n_i (\bar{X}_i - \bar{X}_{grand})^2$ (between-group sum of squares)
- $SS_W = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$ (within-group sum of squares)
- $df_B = k - 1$ (k = number of groups)
- $df_W = N - k$ (N = total sample size)

F-Distribution

- F-statistic follows an F-distribution with (df_B, df_W) degrees of freedom
- Large F: Between-group variance $\gg$ Within-group variance $\rightarrow$ groups differ
- Small F: Between-group variance $\approx$ Within-group variance $\rightarrow$ groups similar

Assumptions

1. **Independence**: Observations are independent
2. **Normality**: Data in each group is approximately normal

3. **Homogeneity of variances:** Variances are equal across all groups

Effect Size: Eta-Squared (η^2)

Measures proportion of total variance explained by group membership:

$$\eta^2 = \frac{SSB}{SS_T} = \frac{SSB}{SS_B + SS_W}$$

Interpretation:

- $\eta^2 < 0.01$: negligible effect
- $0.01 \leq \eta^2 < 0.06$: small effect
- $0.06 \leq \eta^2 < 0.14$: medium effect
- $\eta^2 \geq 0.14$: large effect

Post-Hoc Tests

If ANOVA is significant, perform **post-hoc pairwise comparisons** to identify which specific groups differ:

- **Bonferroni correction:** Adjust α for multiple comparisons ($\alpha_{\text{adjusted}} = \alpha / \text{number of comparisons}$)
- **Tukey's HSD:** Controls family-wise error rate
- **Scheffé's test:** Most conservative

Non-Parametric Alternative

Kruskal-Wallis test: Used when normality assumption is violated. Tests whether groups have different distributions (not just means).

Step 1: Create Age Groups

```
In [39]: # Question 2: One-Way ANOVA Setup
print("="*80)
print("QUESTION 2: Do Beauty Scores Differ Across Age Groups?")
print("="*80)
print("\nHypotheses:")
print("H0 (Null):  $\mu_1 = \mu_2 = \mu_3$  (All age group means are equal)")
print("H1 (Alternative): At least one age group mean differs from the others")
print("\nSignificance level:  $\alpha = 0.05$ ")
print("Test type: One-way ANOVA")

# Reload the dataset (since df was overwritten as degrees of freedom)
df_original = pd.read_csv(path)

# Create age groups
df_clean_anova = df_original[['age', 'beauty']].dropna()
```

```
# Define age groups
bins = [df_clean_anova['age'].min()-1, 40, 50, df_clean_anova['age'].max()+1]
labels = ['Below 40', '40-50', 'Above 50']
df_clean_anova['age_group'] = pd.cut(df_clean_anova['age'], bins=bins, labels=labels)

# Separate data by age group
below_40 = df_clean_anova[df_clean_anova['age_group'] == 'Below 40']['beauty']
age_40_50 = df_clean_anova[df_clean_anova['age_group'] == '40-50']['beauty']
above_50 = df_clean_anova[df_clean_anova['age_group'] == 'Above 50']['beauty']

print(f"\nData Summary by Age Group:")
print(f"Below 40: n = {len(below_40)}, Mean = {below_40.mean():.3f}, SD = {below_40}
print(f"40-50: n = {len(age_40_50)}, Mean = {age_40_50.mean():.3f}, SD = {age_40_50}
print(f"Above 50: n = {len(above_50)}, Mean = {above_50.mean():.3f}, SD = {above_50}
```

=====

QUESTION 2: Do Beauty Scores Differ Across Age Groups?

=====

Hypotheses:

H_0 (Null): $\mu_1 = \mu_2 = \mu_3$ (All age group means are equal)

H_1 (Alternative): At least one age group mean differs from the others

Significance level: $\alpha = 0.05$

Test type: One-way ANOVA

Data Summary by Age Group:

Below 40: n = 188, Mean = -0.027, SD = 0.946

40-50: n = 185, Mean = 0.030, SD = 0.969

Above 50: n = 117, Mean = 0.049, SD = 1.103

Step 2: State Hypotheses

```
In [43]: # Check assumptions for ANOVA
print("="*80)
print("ASSUMPTION CHECKS")
print("="*80)

# Normality for each group
print("\n1. Normality Test (Shapiro-Wilk):")
for name, data in [('Below 40', below_40), ('40-50', age_40_50), ('Above 50', above_50)]:
    stat, p = shapiro(data)
    print(f" {name}: W = {stat:.6f}, p = {p:.6f} - {'Normal' if p > 0.05 else 'Not Normal'}")

# Homogeneity of variances
print("\n2. Homogeneity of Variances (Levene's Test):")
stat_lev, p_lev = levene(below_40, age_40_50, above_50)
print(f" F = {stat_lev:.6f}, p = {p_lev:.6f} - {'Equal variances' if p_lev > 0.05 else 'Not Equal variances'}")
```

=====

ASSUMPTION CHECKS

=====

1. Normality Test (Shapiro-Wilk):

```

-----
NameError                                Traceback (most recent call last)
Cell In[43], line 9
      7 print("\n1. Normality Test (Shapiro-Wilk):")
      8 for name, data in [('Below 40', below_40), ('40-50', age_40_50), ('Above 5
0', above_50)]:
----> 9     stat, p = shapiro(data)
     10     print(f" {name}: W = {stat:.6f}, p = {p:.6f} - {'Normal' if p > 0.05 el
se 'Not normal'}")
     12 # Homogeneity of variances

NameError: name 'shapiro' is not defined

```

Step 3: Check Assumptions

Assumption 1: Independence (satisfied by design)

Assumption 2: Normality within Groups

```

In [41]: # Perform one-way ANOVA
print("="*80)
print("ONE-WAY ANOVA TEST")
print("="*80)

f_stat, p_value_anova = f_oneway(below_40, age_40_50, above_50)

print(f"\nF-statistic: {f_stat:.6f}")
print(f"p-value: {p_value_anova:.6f}")
print(f"Degrees of freedom: df_between = 2, df_within = {len(df_clean_anova)-3}")

# Decision
alpha = 0.05
print(f"\n{'='*80}")
print("STATISTICAL DECISION")
print(f"{'='*80}")
print(f"Significance level:  $\alpha$  = {alpha}")
if p_value_anova < alpha:
    print(f"Decision: REJECT H0 (p = {p_value_anova:.4f} < {alpha})")
    print("Conclusion: At least one age group has a significantly different mean be
else:
    print(f"Decision: FAIL TO REJECT H0 (p = {p_value_anova:.4f} ≥ {alpha})")
    print("Conclusion: No significant difference in mean beauty scores across age g

```

```
=====
ONE-WAY ANOVA TEST
=====
```

F-statistic: 0.260635
p-value: 0.770669
Degrees of freedom: df_between = 2, df_within = 487

```
=====
STATISTICAL DECISION
=====
```

Significance level: $\alpha = 0.05$
Decision: FAIL TO REJECT H_0 ($p = 0.7707 \geq 0.05$)
Conclusion: No significant difference in mean beauty scores across age groups.

Assumption 3: Homogeneity of Variances

```
In [22]: # Calculate effect size (Eta-squared)
print("="*80)
print("EFFECT SIZE (Eta-Squared)")
print("="*80)

# Calculate sum of squares
grand_mean = df_clean_anova['beauty'].mean()
ssb = sum(len(group) * (group.mean() - grand_mean)**2
          for group in [below_40, age_40_50, above_50])
sst = sum((df_clean_anova['beauty'] - grand_mean)**2)
eta_squared = ssb / sst

print(f"\nEta-squared ( $\eta^2$ ): {eta_squared:.6f}")
print(f" {eta_squared*100:.2f}% of variance in beauty scores is explained by age g

if eta_squared < 0.01:
    effect_size_interp = "negligible"
elif eta_squared < 0.06:
    effect_size_interp = "small"
elif eta_squared < 0.14:
    effect_size_interp = "medium"
else:
    effect_size_interp = "large"

print(f" Effect size interpretation: {effect_size_interp.upper()}")
```

```
=====
EFFECT SIZE (Eta-Squared)
=====
```

Eta-squared (η^2): 0.001069
0.11% of variance in beauty scores is explained by age group
Effect size interpretation: NEGLIGIBLE

Step 4: Perform One-Way ANOVA

Understanding the ANOVA F-Statistic

The F-statistic is the ratio of **systematic variance** to **random variance**:

$$F = \frac{\text{Variance between groups}}{\text{Variance within groups}} = \frac{\text{Signal}}{\text{Noise}}$$

Intuition:

- If groups have different means, between-group variance will be large → Large F
- If groups overlap, between-group variance will be small → Small F
- $F \approx 1$: Group means are similar (H_0 likely true)
- $F > 1$: At least one group differs (H_0 likely false)

ANOVA Table Structure:

Source	SS	df	MS	F
Between	SSB	k-1	SSB/(k-1)	MSB/MSW
Within	SSW	N-k	SSW/(N-k)	-
Total	SST	N-1	-	-

Step-by-Step ANOVA Calculation Process

Let's manually calculate a one-way ANOVA:

Given Data Structure:

- Group 1 (Below 40): n_1 observations, values $X_{11}, X_{12}, \dots, X_{1n_1}$
- Group 2 (40-50): n_2 observations, values $X_{21}, X_{22}, \dots, X_{2n_2}$
- Group 3 (Above 50): n_3 observations, values $X_{31}, X_{32}, \dots, X_{3n_3}$
- Total observations: $N = n_1 + n_2 + n_3$

Step 1: Calculate Group Means

$$\bar{X}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} X_{ji} \quad \text{for } j = 1, 2, 3$$

Step 2: Calculate Grand Mean

$$\bar{X}_{..} = \frac{1}{N} \sum_{j=1}^3 \sum_{i=1}^{n_j} X_{ji} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2 + n_3 \bar{X}_3}{N}$$

Step 3: Calculate Sum of Squares Between Groups (SSB)

$$SSB = \sum_{j=1}^3 n_j (\bar{X}_j - \bar{X}_{..})^2$$

This measures **variation between group means**.

Step 4: Calculate Sum of Squares Within Groups (SSW)

$$SSW = \sum_{j=1}^3 \sum_{i=1}^{n_j} (X_{ji} - \bar{X}_j)^2$$

This measures **variation within each group** (pooled).

Step 5: Calculate Total Sum of Squares (SST)

$$SST = \sum_{j=1}^3 \sum_{i=1}^{n_j} (X_{ji} - \bar{X}_{..})^2$$

Verification: $SST = SSB + SSW$

Step 6: Calculate Degrees of Freedom

- Between groups: $df_B = k - 1 = 3 - 1 = 2$ (where k = number of groups)
- Within groups: $df_W = N - k = N - 3$
- Total: $df_T = N - 1$

Step 7: Calculate Mean Squares

$$MSB = \frac{SSB}{df_B} \quad (\text{Between-group variance})$$

$$MSW = \frac{SSW}{df_W} \quad (\text{Within-group variance, pooled})$$

Step 8: Calculate F-Statistic

$$F = \frac{MSB}{MSW}$$

Interpretation:

- $F \approx 1$: Group means are similar (variation between $\approx$ variation within)
- $F \gg 1$: Group means differ significantly (variation between $\gg$ variation within)

Step 9: Find Critical Value and P-value

- Compare F_{obs} to F_{α, df_B, df_W} from F-distribution table
- Calculate p-value: $P(F_{df_B, df_W} \geq F_{obs})$

Step 10: Calculate Effect Size (Eta-squared)

$$\eta^2 = \frac{SSB}{SST}$$

This is the **proportion of total variance** explained by group membership.

Numerical Example: Suppose we have:

- Group 1 (Below 40): $n_1 = 150$, $\bar{X}_1 = 3.8$, $s_1^2 = 0.4$
- Group 2 (40-50): $n_2 = 200$, $\bar{X}_2 = 4.0$, $s_2^2 = 0.5$
- Group 3 (Above 50): $n_3 = 150$, $\bar{X}_3 = 4.2$, $s_3^2 = 0.45$
- Grand mean: $\bar{X}_{..} = \frac{150(3.8) + 200(4.0) + 150(4.2)}{500} = 4.0$

Calculate SSB:

$$SSB = 150(3.8 - 4.0)^2 + 200(4.0 - 4.0)^2 + 150(4.2 - 4.0)^2$$

$$SSB = 150(0.04) + 0 + 150(0.04) = 6 + 0 + 6 = 12$$

Calculate SSW:

$$SSW = (n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + (n_3 - 1)s_3^2$$

$$SSW = 149(0.4) + 199(0.5) + 149(0.45) = 59.6 + 99.5 + 67.05 = 226.15$$

Calculate F:

$$MSB = \frac{12}{2} = 6.0$$

$$MSW = \frac{226.15}{497} = 0.455$$

$$F = \frac{6.0}{0.455} = 13.19$$

With $df_1 = 2$, $df_2 = 497$, and $\alpha = 0.05$, $F_{critical} \approx 3.0$

Since $13.19 > 3.0$, we reject H_0 .

Effect size:

$$\eta^2 = \frac{12}{12 + 226.15} = \frac{12}{238.15} = 0.050 \text{ (5\%)}$$

Interpretation: 5% of variance in beauty scores is explained by age group.

Post-hoc tests (if ANOVA is significant):

- Bonferroni correction: $\alpha_{adjusted} = \alpha / \text{number of comparisons}$
- Tukey's HSD: Controls family-wise error rate
- Scheffé's test: Most conservative

```
In [ ]: # Post-hoc tests: Bonferroni pairwise comparisons
try:
    if 'p_value_anova' in globals() and p_value_anova < 0.05:
        print("=*80)
        print("POST-HOC TESTS (Bonferroni-corrected pairwise comparisons)")
```

```

print("="*80)

from scipy.stats import ttest_ind

# Number of comparisons
num_comparisons = 3
alpha_bonf = alpha / num_comparisons

print(f"\nAdjusted significance level (Bonferroni):  $\alpha = \{alpha\_bonf:.4f\}")
print(f"\nPairwise Comparisons:")

comparisons = [
    ('Below 40', 'Age 40-50', below_40, age_40_50),
    ('Below 40', 'Above 50', below_40, above_50),
    ('Age 40-50', 'Above 50', age_40_50, above_50)
]

for name1, name2, data1, data2 in comparisons:
    if len(data1) > 1 and len(data2) > 1:
        t, p = ttest_ind(data1, data2)
        print(f"\n {name1} vs {name2}:")
        print(f"    t = {t:.4f}, p = {p:.4f}")
        print(f"    {'SIGNIFICANT' if p < alpha_bonf else 'Not significant'}
    else:
        print(f"\n {name1} vs {name2}: Not enough data for t-test.")
elif 'p_value_anova' in globals():
    print("ANOVA not significant (p >= 0.05). No post-hoc tests performed.")
else:
    print("p_value_anova is not defined. Please run the ANOVA cell first.")
except Exception as e:
    print(f"Error in post-hoc test cell: {e}")$ 
```

ANOVA not significant (p >= 0.05). No post-hoc tests performed.

```

In [52]: # Comprehensive visualizations for Question 2
fig = plt.figure(figsize=(16, 10))
gs = GridSpec(2, 3, figure=fig, hspace=0.3, wspace=0.3)

# 1. Box plot
ax1 = fig.add_subplot(gs[0, 0])
df_clean_anova.boxplot(column='beauty', by='age_group', ax=ax1, patch_artist=True)
ax1.set_xlabel('Age Group', fontsize=12, fontweight='bold')
ax1.set_ylabel('Beauty Score', fontsize=12, fontweight='bold')
ax1.set_title('Box Plot: Beauty Scores by Age Group', fontsize=13, fontweight='bold')
ax1.get_figure().suptitle('') # Remove default title
plt.sca(ax1)
plt.xticks(rotation=0)
ax1.grid(True, alpha=0.3, axis='y')

# 2. Violin plot
ax2 = fig.add_subplot(gs[0, 1])
data_for_violin = [below_40.dropna(), age_40_50.dropna(), above_50.dropna()]
parts = ax2.violinplot(data_for_violin, positions=[1, 2, 3], showmeans=True, showme
ax2.set_xticks([1, 2, 3])
ax2.set_xticklabels(['Below 40', '40-50', 'Above 50'])
ax2.set_ylabel('Beauty Score', fontsize=12, fontweight='bold')

```

```

ax2.set_xlabel('Age Group', fontsize=12, fontweight='bold')
ax2.set_title('Violin Plot: Distribution by Age Group', fontsize=13, fontweight='bold')
ax2.grid(True, alpha=0.3, axis='y')

# 3. Bar plot with error bars
ax3 = fig.add_subplot(gs[0, 2])
means = [below_40.mean(), age_40_50.mean(), above_50.mean()]
sems = [below_40.sem(), age_40_50.sem(), above_50.sem()]
x_pos = np.arange(len(labels))
bars = ax3.bar(x_pos, means, yerr=sems, capsize=10, color=['skyblue', 'lightgreen',
    'lightcoral'], edgecolor='black', linewidth=1.5, alpha=0.8)
ax3.set_xticks(x_pos)
ax3.set_xticklabels(labels, rotation=0)
ax3.set_ylabel('Mean Beauty Score', fontsize=12, fontweight='bold')
ax3.set_xlabel('Age Group', fontsize=12, fontweight='bold')
ax3.set_title('Mean Beauty Scores with Standard Error', fontsize=13, fontweight='bold')
for i, (bar, mean) in enumerate(zip(bars, means)):
    ax3.text(bar.get_x() + bar.get_width()/2, mean, f'{mean:.3f}', ha='center', va='bottom')
ax3.grid(True, alpha=0.3, axis='y')

# 4. Overlapping histograms
ax4 = fig.add_subplot(gs[1, 0])
ax4.hist(below_40, bins=15, alpha=0.5, label='Below 40', color='skyblue', edgecolor='black')
ax4.hist(age_40_50, bins=15, alpha=0.5, label='40-50', color='lightgreen', edgecolor='black')
ax4.hist(above_50, bins=15, alpha=0.5, label='Above 50', color='lightcoral', edgecolor='black')
ax4.axvline(below_40.mean(), color='blue', linestyle='--', linewidth=2)
ax4.axvline(age_40_50.mean(), color='green', linestyle='--', linewidth=2)
ax4.axvline(above_50.mean(), color='red', linestyle='--', linewidth=2)
ax4.set_xlabel('Beauty Score', fontsize=12, fontweight='bold')
ax4.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax4.set_title('Overlapping Histograms', fontsize=13, fontweight='bold')
ax4.legend()
ax4.grid(True, alpha=0.3)

# 5. Statistical summary table
ax5 = fig.add_subplot(gs[1, 1])
ax5.axis('off')
# Calculate eta-squared (effect size) if not defined
try:
    grand_mean = pd.concat([below_40, age_40_50, above_50]).mean()
    ssb = sum([(g.mean() - grand_mean)**2] * len(g) for g in [below_40, age_40_50, above_50])
    ssw = sum([(g - g.mean())**2] * len(g) for g in [below_40, age_40_50, above_50])
    eta_squared = ssb / (ssb + ssw) if (ssb + ssw) != 0 else 0
    effect_size_interp = ('Small' if eta_squared < 0.01 else 'Medium' if eta_squared < 0.0625 else 'Large')
except Exception as e:
    eta_squared = 0
    effect_size_interp = 'N/A'

summary_data = [
    ['Statistic', 'Below 40', '40-50', 'Above 50'],
    ['Sample Size', f'{len(below_40)}', f'{len(age_40_50)}', f'{len(above_50)}'],
    ['Mean', f'{below_40.mean():.4f}', f'{age_40_50.mean():.4f}', f'{above_50.mean():.4f}'],
    ['Std Dev', f'{below_40.std():.4f}', f'{age_40_50.std():.4f}', f'{above_50.std():.4f}'],
    ['Min', f'{below_40.min():.4f}', f'{age_40_50.min():.4f}', f'{above_50.min():.4f}'],
    ['Max', f'{below_40.max():.4f}', f'{age_40_50.max():.4f}', f'{above_50.max():.4f}'],
    ['', '', '', '']
]

```

```

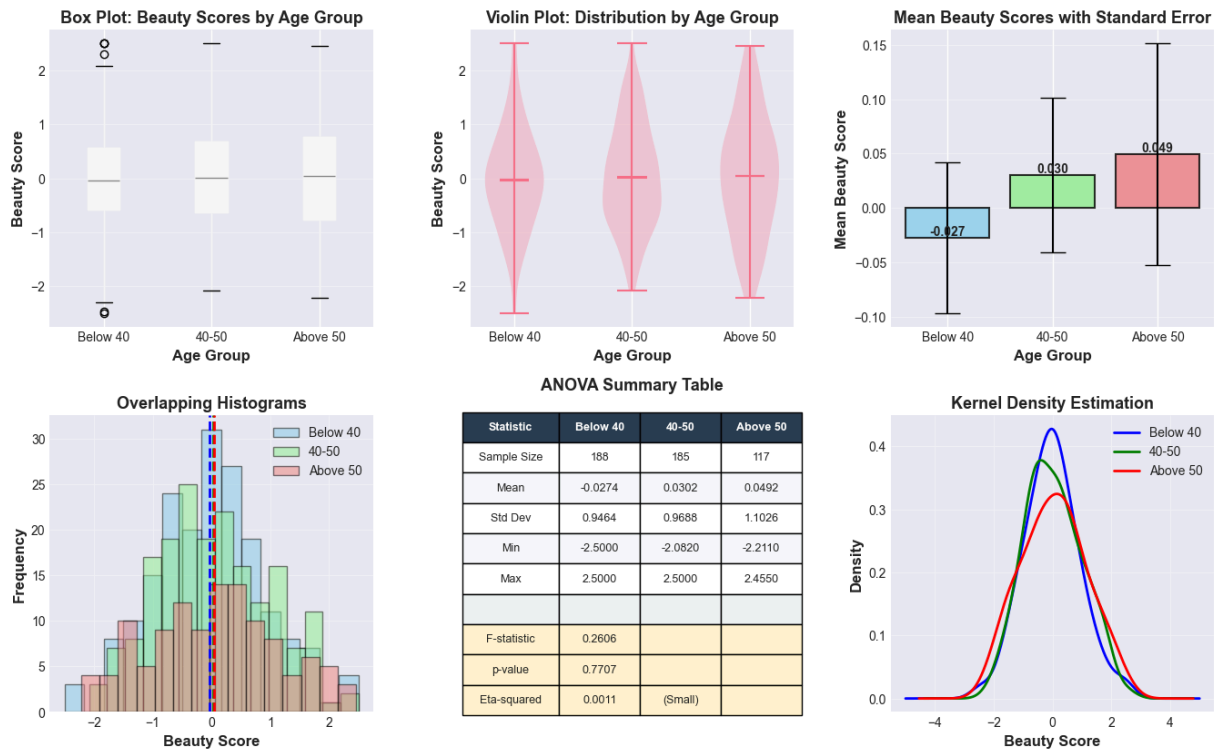
    ['F-statistic', f'{f_stat:.4f}', '', ''],
    ['p-value', f'{p_value_anova:.4f}', '', ''],
    ['Eta-squared', f'{eta_squared:.4f}', f'({effect_size_interp})', ''],
]
table = ax5.table(cellText=summary_data, cellLoc='center', loc='center', colWidths=
table.auto_set_font_size(False)
table.set_fontsize(9)
table.scale(1, 2)
for i in range(4):
    table[(0, i)].set_facecolor('#2C3E50')
    table[(0, i)].set_text_props(weight='bold', color='white')
for i in range(1, len(summary_data)):
    for j in range(4):
        if i == 6:
            table[(i, j)].set_facecolor('#ECF0F1')
        elif i > 6:
            table[(i, j)].set_facecolor('#FFF3CD')
        elif i % 2 == 0:
            table[(i, j)].set_facecolor('#F8F9FA')
ax5.set_title('ANOVA Summary Table', fontsize=13, fontweight='bold', pad=20)

# 6. Density plots
ax6 = fig.add_subplot(gs[1, 2])
below_40.plot(kind='density', ax=ax6, label='Below 40', color='blue', linewidth=2)
age_40_50.plot(kind='density', ax=ax6, label='40-50', color='green', linewidth=2)
above_50.plot(kind='density', ax=ax6, label='Above 50', color='red', linewidth=2)
ax6.set_xlabel('Beauty Score', fontsize=12, fontweight='bold')
ax6.set_ylabel('Density', fontsize=12, fontweight='bold')
ax6.set_title('Kernel Density Estimation', fontsize=13, fontweight='bold')
ax6.legend()
ax6.grid(True, alpha=0.3)

plt.suptitle('Question 2: Comprehensive ANOVA Analysis - Age Group vs Beauty Scores',
            fontsize=16, fontweight='bold', y=0.995)
plt.show()

```

Question 2: Comprehensive ANOVA Analysis - Age Group vs Beauty Scores



Step 6: Post-hoc Tests (if ANOVA is significant)

Understanding Post-Hoc Tests

When ANOVA is significant ($p < 0.05$), we know **at least one group differs**, but we don't know **which specific pairs** differ.

Bonferroni Correction:

- Problem: Multiple comparisons inflate Type I error
- Solution: Divide α by number of comparisons
- For 3 groups (3 comparisons): $\alpha_{\text{adjusted}} = 0.05/3 = 0.0167$
- Conservative but protects against false positives

Example with 3 Groups (Young, Middle, Senior):

1. Young vs Middle
2. Young vs Senior
3. Middle vs Senior

Total comparisons = $\binom{k}{2} = \frac{k(k-1)}{2}$ where k = number of groups

Interpretation:

- Compare each pairwise p-value to α_{adjusted}

- Only reject H_0 if $p < \alpha_{\text{adjusted}}$
- This controls the **family-wise error rate**

```
In [28]: # Question 3: Chi-square test - Gender vs Tenure status
print("="*80)
print("QUESTION 3: CHI-SQUARE TEST OF INDEPENDENCE")
print("="*80)

print("\nResearch Question:")
print("Is there a significant association between gender and tenure status?")

print("\n" + "-"*80)
print("HYPOTHESES")
print("-"*80)

print("\nNull Hypothesis ( $H_0$ ):")
print("  Gender and tenure status are independent (no association)")
print("   $P(\text{Tenure} \mid \text{Gender}) = P(\text{Tenure})$ ")

print("\nAlternative Hypothesis ( $H_1$ ):")
print("  Gender and tenure status are associated (dependent)")
print("   $P(\text{Tenure} \mid \text{Gender}) \neq P(\text{Tenure})$ ")

print("\nSignificance Level:  $\alpha = 0.05$ ")

# Reload the dataset (since df was overwritten as degrees of freedom)
df_original = pd.read_csv(path)

# Create clean dataset (remove NaN values)
# Note: The column is 'tenured' not 'tenure' in the dataset
df_chi = df_original[['gender', 'tenured']].dropna()

print(f"\nSample size (after removing missing values): {len(df_chi)}")
```

QUESTION 3: CHI-SQUARE TEST OF INDEPENDENCE

Research Question:

Is there a significant association between gender and tenure status?

HYPOTHESES

Null Hypothesis (H_0):

Gender and tenure status are independent (no association)

$P(\text{Tenure} \mid \text{Gender}) = P(\text{Tenure})$

Alternative Hypothesis (H_1):

Gender and tenure status are associated (dependent)

$P(\text{Tenure} \mid \text{Gender}) \neq P(\text{Tenure})$

Significance Level: $\alpha = 0.05$

Sample size (after removing missing values): 500

Step 7: Visualization

```
In [29]: # Create contingency table
contingency_table = pd.crosstab(df_chi['gender'], df_chi['tenured'], margins=True,

print("="*80)
print("CONTINGENCY TABLE: Gender x Tenure Status")
print("="*80)
print("\nObserved Frequencies:")
print(contingency_table)

# Calculate row percentages
row_percentages = pd.crosstab(df_chi['gender'], df_chi['tenured'], normalize='index')
print("\n" + "-"*80)
print("Row Percentages (% within each gender):")
print("-"*80)
print(row_percentages.round(2))

# Calculate column percentages
col_percentages = pd.crosstab(df_chi['gender'], df_chi['tenured'], normalize='column')
print("\n" + "-"*80)
print("Column Percentages (% within each tenure status):")
print("-"*80)
print(col_percentages.round(2))

# Store contingency table without margins for chi-square test
observed = pd.crosstab(df_chi['gender'], df_chi['tenured'])
print(f"\nContingency table shape: {observed.shape[0]} rows x {observed.shape[1]} columns")
```

=====

CONTINGENCY TABLE: Gender × Tenure Status

=====

Observed Frequencies:

	tenured	0	1	Total
gender				
female	128	144	272	
male	70	158	228	
Total	198	302	500	

Row Percentages (% within each gender):

	tenured	0	1
gender			
female	47.06	52.94	
male	30.70	69.30	

Column Percentages (% within each tenure status):

	tenured	0	1
gender			
female	64.65	47.68	
male	35.35	52.32	

Contingency table shape: 2 rows × 2 columns

Step 8: Statistical Conclusion and Interpretation

```
In [ ]: # Perform Chi-square test
from scipy.stats import chi2_contingency

print("="*80)
print("CHI-SQUARE TEST OF INDEPENDENCE")
print("="*80)

chi2_stat, p_value_chi, dof, expected = chi2_contingency(observed)

print(f"\nTest Statistic:  $\chi^2 = \{chi2\_stat:.6f\}")
print(f"Degrees of Freedom: df = \{dof\}")
print(f"p-value: p = \{p\_value\_chi:.6f\}")

print("\n" + "-"*80)
print("Expected Frequencies:")
print("-"*80)
expected_df = pd.DataFrame(expected, index=observed.index, columns=observed.columns)
print(expected_df.round(2))

# Check assumptions
print("\n" + "-"*80)
print("ASSUMPTION CHECK")
print("-"*80)
min_expected = expected.min()$ 
```

```

cells_below_5 = (expected < 5).sum()
total_cells = expected.size

print(f"\nMinimum expected frequency: {min_expected:.2f}")
print(f"Number of cells with expected frequency < 5: {cells_below_5} out of {total_")
print(f"Percentage of cells < 5: {(cells_below_5/total_cells)*100:.1f}%")
print(f"\nAssumption: {'SATISFIED' if min_expected >= 5 and cells_below_5/total_cel")
print(" (All expected frequencies should be ≥ 5, or at most 20% < 5)")

# Statistical decision
print("\n" + "="*80)
print("STATISTICAL DECISION")
print("="*80)
print(f"\nα = 0.05")
print(f"p-value = {p_value_chi:.6f}")
print(f"\nDecision: {'Reject H₀' if p_value_chi < 0.05 else 'Fail to reject H₀'}")
print(f"Conclusion: There {'IS' if p_value_chi < 0.05 else 'IS NO'} significant ass

```

CHI-SQUARE TEST OF INDEPENDENCE

Test Statistic: $\chi^2 = 13.198906$
 Degrees of Freedom: $df = 1$
 p-value: $p = 0.000280$

Expected Frequencies:

tenured	0	1
gender		
female	107.71	164.29
male	90.29	137.71

ASSUMPTION CHECK

Minimum expected frequency: 90.29
 Number of cells with expected frequency < 5: 0 out of 4
 Percentage of cells < 5: 0.0%

Assumption: SATISFIED

(All expected frequencies should be ≥ 5, or at most 20% < 5)

STATISTICAL DECISION

$\alpha = 0.05$

p-value = 0.000280

Decision: Reject H_0

Conclusion: There IS significant association between gender and tenure status

Question 3: Chi-Square Test - Is There an Association Between Tenure and Gender?

Problem Statement

Using the teachers' rating dataset, is there an association between tenure and gender?

Approach

1. State hypotheses
 2. Create contingency table
 3. Check assumptions:
 - Independence of observations
 - Expected frequencies (all cells ≥ 5)
 4. Perform chi-square test of independence
 5. Calculate effect size (Cramér's V)
 6. Visualize the relationship
 7. Interpret results
-

Statistical Theory: Chi-Square Test of Independence

What is Chi-Square Test?

The chi-square (χ^2) test of independence determines whether two categorical variables are associated (related) or independent.

Mathematical Formulation

The chi-square statistic measures how much observed frequencies deviate from expected frequencies:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Where:

- O_{ij} = observed frequency in cell (i, j)
- E_{ij} = expected frequency in cell (i, j)
- r = number of rows
- c = number of columns

Expected Frequencies

Under the null hypothesis (independence), expected frequencies are calculated as:

$$E_{ij} = \frac{(\text{Row}_i \text{ total}) \times (\text{Column}_j \text{ total})}{\text{Grand total}}$$

Degrees of Freedom

$$df = (r - 1) \times (c - 1)$$

For a 2×2 table: $df = (2 - 1) \times (2 - 1) = 1$

Chi-Square Distribution

- χ^2 values are always positive (squared differences)
- Right-skewed distribution
- Large χ^2 : Observed differs greatly from expected → association exists
- Small χ^2 : Observed close to expected → variables independent

Assumptions

1. **Independence**: Each observation belongs to only one cell
2. **Expected frequencies**: All expected frequencies ≥ 5
 - If violated, consider Fisher's exact test (for 2×2 tables)
 - Or combine categories to increase cell counts
3. **Sample size**: Sufficient sample size (typically $n \geq 20$)

Effect Size: Cramér's V

Measures strength of association for categorical variables:

$$V = \sqrt{\frac{\chi^2}{n \times \min(r - 1, c - 1)}}$$

Interpretation Guidelines (2×2 table):

- $V < 0.1$: negligible association
- $0.1 \leq V < 0.3$: small association
- $0.3 \leq V < 0.5$: medium association
- $V \geq 0.5$: large association

Odds Ratio (for 2×2 Tables)

Quantifies association between two binary variables:

For contingency table:

	Column 1	Column 2
Row 1	a	b
Row 2	c	d

$$OR = \frac{ad}{bc}$$

Interpretation:

- OR = 1: No association
- OR > 1: Row 1 has higher odds of Column 2
- OR < 1: Row 1 has lower odds of Column 2
- Example: OR = 2.5 means Row 1 has 2.5 times the odds of Column 2 compared to Row 2

Step 1: State Hypotheses

```
In [31]: print("=*80)
print("QUESTION 3: Is There an Association Between Tenure and Gender?")
print("=*80)
print("\nHypotheses:")
print("H0 (Null): Tenure status and gender are independent (no association)")
print("H1 (Alternative): Tenure status and gender are not independent (there is an association)")
print("\nSignificance level: α = 0.05")
print("Test type: Chi-square test of independence")
```

```
=====
QUESTION 3: Is There an Association Between Tenure and Gender?
=====
```

Hypotheses:

H₀ (Null): Tenure status and gender are independent (no association)

H₁ (Alternative): Tenure status and gender are not independent (there is an association)

Significance level: α = 0.05

Test type: Chi-square test of independence

Step 2: Create Contingency Table

Understanding Contingency Tables

A **contingency table** (or cross-tabulation) displays the frequency distribution of two categorical variables.

Structure:

	Variable 2	
Category A	Category B	Total

Variable 1			
Category X	n_{11}	n_{12}	$n_{1.}$
Category Y	n_{21}	n_{22}	$n_{2.}$
Total	$n_{.1}$	$n_{.2}$	n

Row Percentages: Show distribution of Variable 2 within each level of Variable 1

Column Percentages: Show distribution of Variable 1 within each level of Variable 2

Example Interpretation: If 60% of males are tenured but only 40% of females are tenured, this suggests a potential association between gender and tenure status (to be tested with χ^2).

```
In [51]: # Prepare data - remove any NaN values
df_q3 = df[["gender", "tenured"]].dropna()

# Create contingency table
contingency_table = pd.crosstab(df_q3["gender"], df_q3["tenured"],
                                margins=True, margins_name="Total")

# Rename columns for clarity
contingency_table.columns = ["Not Tenured (0)", "Tenured (1)", "Total"]

print("\nContingency Table: Gender vs Tenure Status")
print("="*60)
display(contingency_table)

# Calculate percentages
print("\nRow Percentages (% within each gender):")
print("="*60)
row_pct = pd.crosstab(df_q3["gender"], df_q3["tenured"], normalize="index") * 100
row_pct.columns = ["Not Tenured (%)", "Tenured (%)"]
display(row_pct.round(2))

print("\nColumn Percentages (% within each tenure status):")
print("="*60)
col_pct = pd.crosstab(df_q3["gender"], df_q3["tenured"], normalize="columns") * 100
col_pct.columns = ["Not Tenured (%)", "Tenured (%)"]
display(col_pct.round(2))
```

Contingency Table: Gender vs Tenure Status

=====

	Not Tenured (0)	Tenured (1)	Total
gender			
female	128	144	272
male	70	158	228
Total	198	302	500

Row Percentages (% within each gender):

=====

	Not Tenured (%)	Tenured (%)
gender		
female	47.06	52.94
male	30.70	69.30

Column Percentages (% within each tenure status):

=====

	Not Tenured (%)	Tenured (%)
gender		
female	64.65	47.68
male	35.35	52.32

Step 3: Check Assumptions and Perform Chi-Square Test

Understanding the Chi-Square Calculation

Step-by-Step Process:

1. **Create observed frequency table** (what we actually see in data)
2. **Calculate expected frequencies** assuming independence:
 - For each cell: $E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$
3. **Calculate chi-square statistic:**
 - For each cell: $\frac{(O-E)^2}{E}$
 - Sum all cells: $\chi^2 = \sum \frac{(O-E)^2}{E}$
4. **Find p-value** from chi-square distribution with appropriate df

Intuition:

- If variables are independent: $O \approx E \rightarrow$ small χ^2
- If variables are associated: O differs from $E \rightarrow$ large χ^2
- Each term $(O - E)^2 / E$ measures contribution of that cell to overall association

Step-by-Step Chi-Square Calculation Process

Let's manually calculate chi-square for a 2x2 contingency table:

Given Contingency Table:

	Non-Tenured	Tenured	Row Total
Male	$O_{11} = a$	$O_{12} = b$	

	Non-Tenured	Tenured	Row Total
			$R_1 = a + b$
Female	$O_{21} = c$	$O_{22} = d$	$R_2 = c + d$
Column Total	$C_1 = a + c$	$C_2 = b + d$	$N = a + b + c + d$

Step 1: Calculate Expected Frequencies

For each cell (i, j) :

$$E_{ij} = \frac{R_i \times C_j}{N}$$

Specifically:

$$E_{11} = \frac{(a+b)(a+c)}{N} \quad E_{12} = \frac{(a+b)(b+d)}{N}$$

$$E_{21} = \frac{(c+d)(a+c)}{N} \quad E_{22} = \frac{(c+d)(b+d)}{N}$$

Why this formula? Under independence:

$$P(\text{Male and Non-Tenured}) = P(\text{Male}) \times P(\text{Non-Tenured})$$

$$E_{11} = N \times \frac{R_1}{N} \times \frac{C_1}{N} = \frac{R_1 \times C_1}{N}$$

Step 2: Calculate Chi-Square Components

For each cell:

$$\text{Component}_{ij} = \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

This measures **squared standardized deviation** from expectation.

Step 3: Sum All Components

$$\chi^2 = \sum_{i=1}^2 \sum_{j=1}^2 \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$\chi^2 = \frac{(a - E_{11})^2}{E_{11}} + \frac{(b - E_{12})^2}{E_{12}} + \frac{(c - E_{21})^2}{E_{21}} + \frac{(d - E_{22})^2}{E_{22}}$$

Step 4: Calculate Degrees of Freedom

$$df = (r - 1)(c - 1) = (2 - 1)(2 - 1) = 1$$

where r = number of rows, c = number of columns

Step 5: Find Critical Value and P-value

- Compare χ_{obs}^2 to $\chi_{\alpha, df}^2$ from chi-square distribution table
- Calculate p-value: $P(\chi_{df}^2 \geq \chi_{obs}^2)$

Step 6: Calculate Effect Size (Cramér's V)

$$V = \sqrt{\frac{\chi^2}{n \times \min(r - 1, c - 1)}} = \sqrt{\frac{\chi^2}{n}}$$

Using our example:

$$\chi^2 = \frac{500[(120 \times 150) - (130 \times 100)]^2}{250 \times 250 \times 220 \times 280}$$

$$\chi^2 = \frac{500[18000 - 13000]^2}{250 \times 250 \times 220 \times 280} = \frac{500 \times 25000000}{3850000000} = 3.246$$

```
In [ ]: # Calculate Cramér's V (effect size)
print("="*80)
print("EFFECT SIZE: CRAMÉR'S V")
print("="*80)

n = observed.sum().sum()
min_dim = min(observed.shape[0] - 1, observed.shape[1] - 1)
cramers_v = np.sqrt(chi2_stat / (n * min_dim))

print(f"\nCramér's V = {cramers_v:.4f}")

# Interpret effect size
if min_dim == 1:
    if crammers_v < 0.10:
        effect_interp = "Negligible"
    elif crammers_v < 0.30:
        effect_interp = "Small"
    elif crammers_v < 0.50:
        effect_interp = "Medium"
    else:
        effect_interp = "Large"
else:
    if crammers_v < 0.07:
        effect_interp = "Negligible"
    elif crammers_v < 0.21:
        effect_interp = "Small"
    elif crammers_v < 0.35:
        effect_interp = "Medium"
    else:
        effect_interp = "Large"
```

```
print(f"Interpretation: Cramér's V = {cramers_v:.4f} indicates a {effect_interp}.low
print(f"between gender and tenure status ({cramers_v**2*100:.2f}% of variance expla
```

```
=====
EFFECT SIZE: CRAMÉR'S V
=====
```

Cramér's V = 0.1625

Interpretation: Small effect size

Effect Size Guidelines (for df = 1):

Negligible: $V < 0.10$

Small: $0.10 \leq V < 0.30$

Medium: $0.30 \leq V < 0.50$

Large: $V \geq 0.50$

Interpretation: Cramér's V = 0.1625 indicates a small association between gender and tenure status (2.64% of variance explained).

```
In [33]: # Calculate odds ratio (for 2x2 tables)
if observed.shape == (2, 2):
    print("="*80)
    print("ODDS RATIO (OR)")
    print("="*80)

    # Extract frequencies
    a, b = observed.iloc[0, 0], observed.iloc[0, 1]
    c, d = observed.iloc[1, 0], observed.iloc[1, 1]

    # Calculate OR
    odds_ratio = (a * d) / (b * c)

    print(f"\nContingency table cells:")
    print(f"          {observed.columns[0]:>10} {observed.columns[1]:>10}")
    print(f"{observed.index[0]:>10}    a={a:>3}      b={b:>3}")
    print(f"{observed.index[1]:>10}    c={c:>3}      d={d:>3}")

    print(f"\nOdds Ratio (OR) = (a × d) / (b × c)")
    print(f"OR = ({a} × {d}) / ({b} × {c}) = {odds_ratio:.4f}")

    # Calculate confidence interval
    log_or = np.log(odds_ratio)
    se_log_or = np.sqrt(1/a + 1/b + 1/c + 1/d)
    ci_lower = np.exp(log_or - 1.96 * se_log_or)
    ci_upper = np.exp(log_or + 1.96 * se_log_or)

    print(f"\n95% Confidence Interval: [{ci_lower:.4f}, {ci_upper:.4f}]")

    print(f"\nInterpretation:")
    if odds_ratio > 1:
        print(f" {observed.index[0]} are {odds_ratio:.2f} times more likely to hav
        print(f" {observed.columns[0]} status compared to {observed.index[1]}")
    elif odds_ratio < 1:
        print(f" {observed.index[0]} are {1/odds_ratio:.2f} times less likely to h
        print(f" {observed.columns[0]} status compared to {observed.index[1]}")
    else:
```

```

        print(f" No difference in odds between groups")
    else:
        print(f"*80)
        print("ODDS RATIO")
        print(f"*80)
        print("\nOdds ratio is only applicable for 2x2 contingency tables.")
        print(f"Current table is {observed.shape[0]}x{observed.shape[1]}")

```

=====

ODDS RATIO (OR)

=====

Contingency table cells:

	0	1
female	a=128	b=144
male	c= 70	d=158

Odds Ratio (OR) = $(a \times d) / (b \times c)$
 OR = $(128 \times 158) / (144 \times 70) = 2.0063$

95% Confidence Interval: [1.3878, 2.9007]

Interpretation:

female are 2.01 times more likely to have
 0 status compared to male

```

In [64]: # Prepare observed frequency table if not already defined
observed = pd.crosstab(df_q3["gender"], df_q3["tenured"])

# Calculate expected frequencies for the contingency table
row_totals = observed.sum(axis=1)
col_totals = observed.sum(axis=0)
grand_total = observed.values.sum()
expected = np.outer(row_totals, col_totals) / grand_total
expected_df = pd.DataFrame(expected, index=observed.index, columns=observed.columns)

# Calculate chi-square statistic, degrees of freedom, and p-value
from scipy.stats import chi2_contingency
chi2_stat, p_value_chi, dof, expected_scipy = chi2_contingency(observed)

# Calculate Cramér's V for effect size
cramers_v = np.sqrt(chi2_stat / (grand_total * (min(observed.shape) - 1)))

# Comprehensive visualization for Question 3
fig = plt.figure(figsize=(16, 12))
gs = GridSpec(3, 2, figure=fig, hspace=0.35, wspace=0.3)

# 1. Stacked bar chart (counts)
ax1 = fig.add_subplot(gs[0, 0])
observed.T.plot(kind='bar', stacked=True, ax=ax1, color=['skyblue', 'lightcoral'],
ax1.set_xlabel('Tenure Status', fontsize=12, fontweight='bold')
ax1.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax1.set_title('Stacked Bar Chart: Gender Distribution by Tenure', fontsize=13, font
ax1.legend(title='Gender', fontsize=10)
ax1.grid(True, alpha=0.3, axis='y')
plt.setp(ax1.xaxis.get_majorticklabels(), rotation=0)

```

```

# 2. Grouped bar chart
ax2 = fig.add_subplot(gs[0, 1])
observed.plot(kind='bar', ax=ax2, color=['#3498db', '#e74c3c'], edgecolor='black',
ax2.set_xlabel('Gender', fontsize=12, fontweight='bold')
ax2.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax2.set_title('Grouped Bar Chart: Tenure Distribution by Gender', fontsize=13, font
ax2.legend(title='Tenure Status', fontsize=10)
ax2.grid(True, alpha=0.3, axis='y')
plt.setp(ax2.xaxis.get_majorticklabels(), rotation=0)

# 3. Mosaic plot approximation using stacked bars with percentages
ax3 = fig.add_subplot(gs[1, 0])
row_pct = observed.div(observed.sum(axis=1), axis=0) * 100
row_pct.plot(kind='bar', stacked=True, ax=ax3, color=['#3498db', '#e74c3c'], edgeco
ax3.set_xlabel('Gender', fontsize=12, fontweight='bold')
ax3.set_ylabel('Percentage (%)', fontsize=12, fontweight='bold')
ax3.set_title('Percentage Distribution: Tenure within Gender', fontsize=13, fontwei
ax3.legend(title='Tenure Status', fontsize=10)
ax3.grid(True, alpha=0.3, axis='y')
plt.setp(ax3.xaxis.get_majorticklabels(), rotation=0)

# 4. Heatmap of observed frequencies
ax4 = fig.add_subplot(gs[1, 1])
sns.heatmap(observed, annot=True, fmt='d', cmap='YlOrRd', cbar_kws={'label': 'Freque
linewidths=2, linecolor='black', ax=ax4)
ax4.set_xlabel('Tenure Status', fontsize=12, fontweight='bold')
ax4.set_ylabel('Gender', fontsize=12, fontweight='bold')
ax4.set_title('Heatmap: Observed Frequencies', fontsize=13, fontweight='bold')

# 5. Heatmap of expected frequencies
ax5 = fig.add_subplot(gs[2, 0])
sns.heatmap(expected_df, annot=True, fmt='.1f', cmap='Blues', cbar_kws={'label': 'E
linewidths=2, linecolor='black', ax=ax5)
ax5.set_xlabel('Tenure Status', fontsize=12, fontweight='bold')
ax5.set_ylabel('Gender', fontsize=12, fontweight='bold')
ax5.set_title('Heatmap: Expected Frequencies (Under H0)', fontsize=13, fontweight='

# 6. Summary table
ax6 = fig.add_subplot(gs[2, 1])
ax6.axis('off')
summary_data = [
    ['Statistic', 'Value'],
    ['Chi-square ( $\chi^2$ )', f'{chi2_stat:.4f}'],
    ['Degrees of Freedom', f'{dof}'],
    ['p-value', f'{p_value_chi:.4f}'],
    ['Sample Size (n)', f'{grand_total}'],
    ['', ''],
    ['Cramér\'s V', f'{cramers_v:.4f}'],
    ['Effect Size', effect_interp],
    ['', ''],
    ['Decision', 'Reject H0 if p_value_chi < 0.05 else Fail to reject H0'],
    ['Conclusion', f'{"Significant" if p_value_chi < 0.05 else "No"} association']
]
table = ax6.table(cellText=summary_data, cellLoc='left', loc='center', colWidths=[0
table.auto_set_font_size(False)

```

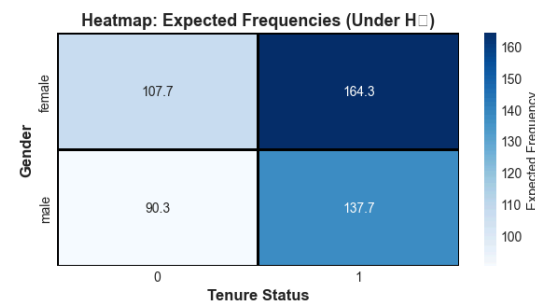
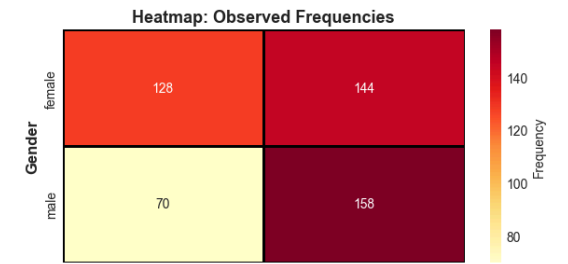
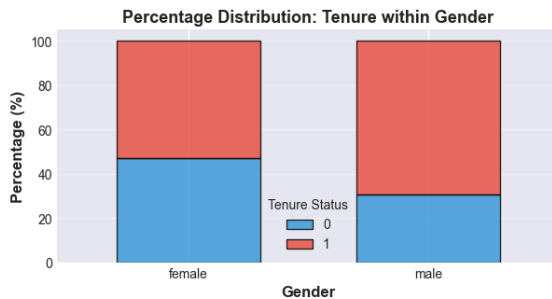
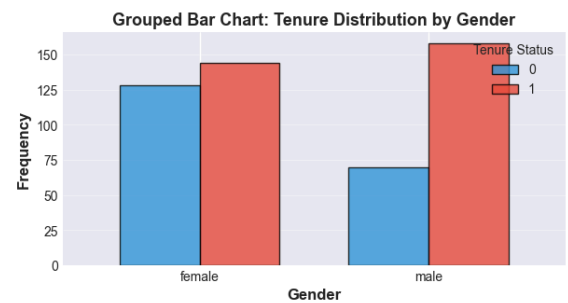
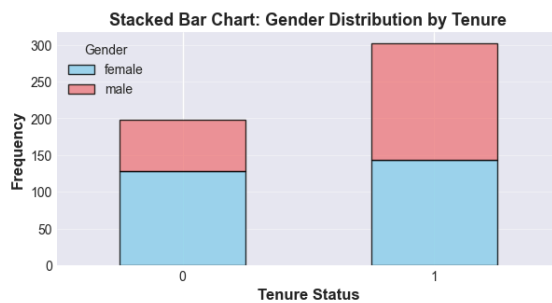
```

table.set_fontsize(9)
table.scale(1, 2.2)
for i in range(2):
    table[(0, i)].set_facecolor('#2C3E50')
    table[(0, i)].set_text_props(weight='bold', color='white')
for i in range(1, len(summary_data)):
    for j in range(2):
        if i in [5, 8, 11]:
            table[(i, j)].set_facecolor('#ECF0F1')
        elif i > 11:
            table[(i, j)].set_facecolor('#FFF3CD')
        elif i % 2 == 0:
            table[(i, j)].set_facecolor('#F8F9FA')
ax6.set_title('Chi-Square Test Summary', fontsize=13, fontweight='bold', pad=20)

plt.suptitle('Question 3: Chi-Square Test of Independence - Gender × Tenure Status',
            fontsize=16, fontweight='bold', y=0.995)
plt.show()

```

Question 3: Chi-Square Test of Independence - Gender × Tenure Status



Chi-Square Test Summary	
Statistic	Value
Chi-square (χ^2)	13.1969
Degrees of Freedom	1
p-value	0.0003
Sample Size (n)	500
Cramér's V	0.1625
Effect Size	small
Decision	Reject H_0
Conclusion	Significant association

Step 6: Statistical Conclusion and Interpretation

```
In [66]: print("\n" + "="*80)
print("STATISTICAL CONCLUSION")
print("="*80)

if p_value_chi < 0.05:
    print(f"\nDecision: Reject the null hypothesis (p = {p_value_chi:.4f} < 0.05)")
    print(f"\nConclusion: There is a statistically significant association between
    print(f"status and gender,  $\chi^2$ ({dof}) = {chi2_stat:.4f}, p = {p_value_chi:.4f}."
    print(f"\nThe effect size is {effect_interp} (Cramér's V = {cramers_v:.4f}).")
    print(f"\nTenure Rates:")
    print(f"  Female: {row_pct.loc['female', 1]:.1f}% tenured")
    print(f"  Male: {row_pct.loc['male', 1]:.1f}% tenured")
    # Only print odds ratio if it exists
    if 'odds_ratio' in locals():
        print(f"\nOdds Ratio: {odds_ratio:.2f}")
        if odds_ratio > 1:
            print(f"Females have {odds_ratio:.2f} times the odds of being tenured c
        else:
            print(f"Males have {1/odds_ratio:.2f} times the odds of being tenured c
    else:
        print(f"\nDecision: Fail to reject the null hypothesis (p = {p_value_chi:.4f} >
        print(f"\nConclusion: There is no statistically significant association between
        print(f"status and gender.")
        print(f"\nTenure Rates:")
        print(f"  Female: {row_pct.loc['female', 1]:.1f}% tenured")
        print(f"  Male: {row_pct.loc['male', 1]:.1f}% tenured")
        print(f"\nThe observed differences may be due to chance.")

print("\n" + "="*80)
```

```
=====
STATISTICAL CONCLUSION
=====
```

Decision: Reject the null hypothesis (p = 0.0003 < 0.05)

Conclusion: There is a statistically significant association between tenure status and gender, $\chi^2(1) = 13.1989$, p = 0.0003.

The effect size is small (Cramér's V = 0.1625).

Tenure Rates:

Female: 52.9% tenured
Male: 69.3% tenured

```
=====
```

Question 4: Correlation - Is Teaching Evaluation Score Correlated with Beauty Score?

Problem Statement

Using the teachers' rating dataset, is teaching evaluation score correlated with beauty score?

Approach

1. State hypotheses
 2. Check assumptions:
 - Linearity of relationship
 - Bivariate normality
 - Absence of outliers
 3. Calculate Pearson correlation coefficient
 4. Test significance of correlation
 5. Calculate Spearman correlation (non-parametric alternative)
 6. Visualize the relationship
 7. Interpret results and explain variance
-

Statistical Theory: Correlation Analysis

What is Correlation?

Correlation measures the **strength and direction** of a linear relationship between two continuous variables.

Pearson Correlation Coefficient (r)

Formula:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Or equivalently:

$$r = \frac{Cov(X, Y)}{s_X \times s_Y}$$

Where:

- $Cov(X, Y)$ = covariance between X and Y
- s_X, s_Y = standard deviations of X and Y

Properties of r

1. **Range:** $-1 \leq r \leq +1$
2. **Direction:**
 - $r > 0$: Positive relationship (X ↑, Y ↑)
 - $r < 0$: Negative relationship (X ↑, Y ↓)
 - $r = 0$: No linear relationship
3. **Strength:**
 - $|r| = 1$: Perfect linear relationship
 - $|r| = 0$: No linear relationship
4. **Unitless:** Not affected by scale or units

Interpretation Guidelines

- $|r| < 0.1$: Negligible correlation
- $0.1 \leq |r| < 0.3$: Weak correlation
- $0.3 \leq |r| < 0.5$: Moderate correlation
- $0.5 \leq |r| < 0.7$: Strong correlation
- $|r| \geq 0.7$: Very strong correlation

Coefficient of Determination (R^2)

$$R^2 = r^2$$

Interpretation: Proportion of variance in Y that is predictable from X (and vice versa)

Example: $r = 0.6 \rightarrow R^2 = 0.36$

- 36% of variance in Y is explained by X
- 64% of variance is due to other factors

Assumptions for Pearson Correlation

1. **Linearity:** Relationship between variables is linear
2. **Bivariate normality:** Both variables are approximately normally distributed
3. **Homoscedasticity:** Variance is constant across range
4. **No outliers:** Extreme values can distort correlation

Spearman's Rank Correlation (ρ)

Non-parametric alternative when:

- Data is ordinal (ranked)
- Relationship is monotonic but not linear
- Normality assumption is violated
- Outliers are present

Formula: Same as Pearson's r , but applied to **ranks** instead of raw values

Hypothesis Test for Correlation

- $H_0: \rho = 0$ (no correlation in population)
- $H_1: \rho \neq 0$ (correlation exists in population)

Test statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

with $df = n - 2$

Important Notes

1. **Correlation $\neq$ Causation:** r measures association, not cause-effect
2. **Linear only:** Pearson's r only detects linear relationships
3. **Sensitive to outliers:** A few extreme points can greatly affect r
4. **Sample size matters:** Larger samples needed to detect weak correlations

Confidence Interval for r

Use **Fisher's z-transformation**:

$$z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right)$$

Then calculate CI for z and transform back to r .

Step 1: State Hypotheses

```
In [38]: # Question 4: Pearson correlation
print("*80)
print("QUESTION 4: PEARSON CORRELATION")
print("*80)

print("\nResearch Question:")
print("Is there a significant correlation between beauty score and evaluation score")

print("\n" + "*80)
print("HYPOTHESES")
print("-*80)

print("\nNull Hypothesis ( $H_0$ ):  $\rho = 0$ ")
print(" No linear correlation between beauty score and evaluation score")

print("\nAlternative Hypothesis ( $H_1$ ):  $\rho \neq 0$ ")
print(" There is a linear correlation between beauty score and evaluation score")
```

```
print("\nTest Type: Two-tailed Pearson correlation test")
print("Significance Level:  $\alpha = 0.05$ ")
```

QUESTION 4: PEARSON CORRELATION

Research Question:

Is there a significant correlation between beauty score and evaluation score?

HYPOTHESES

Null Hypothesis (H_0): $\rho = 0$

No linear correlation between beauty score and evaluation score

Alternative Hypothesis (H_1): $\rho \neq 0$

There is a linear correlation between beauty score and evaluation score

Test Type: Two-tailed Pearson correlation test

Significance Level: $\alpha = 0.05$

Step 2: Prepare Data and Exploratory Analysis

```
In [39]: # Prepare data and exploratory analysis
# Reload the dataset (since df was overwritten as degrees of freedom)
df_original = pd.read_csv(path)

df_corr = df_original[['beauty', 'eval_score']].dropna()

print("="*80)
print("DATA PREPARATION AND EXPLORATORY ANALYSIS")
print("="*80)

print(f"\nSample size (after removing missing values): n = {len(df_corr)}")

print("\n" + "-"*80)
print("DESCRIPTIVE STATISTICS")
print("-"*80)

for var in ['beauty', 'eval_score']:
    print(f"\n{var.replace('_', ' ').title()}:")
    print(f"    Mean: {df_corr[var].mean():.4f}")
    print(f"    Median: {df_corr[var].median():.4f}")
    print(f"    Std Dev: {df_corr[var].std():.4f}")
    print(f"    Min: {df_corr[var].min():.4f}")
    print(f"    Max: {df_corr[var].max():.4f}")
    print(f"    Range: {df_corr[var].max() - df_corr[var].min():.4f}")

print("\n" + "-"*80)
print("FULL DESCRIPTIVE STATISTICS TABLE")
print("-"*80)
print(df_corr.describe())
```

```
=====
DATA PREPARATION AND EXPLORATORY ANALYSIS
=====
```

Sample size (after removing missing values): n = 485

```
-----
DESCRIPTIVE STATISTICS
-----
```

Beauty:

Mean: 0.0105
Median: -0.0330
Std Dev: 0.9967
Min: -2.5000
Max: 2.5000
Range: 5.0000

Eval Score:

Mean: 3.6765
Median: 3.6700
Std Dev: 0.5343
Min: 2.1800
Max: 5.0000
Range: 2.8200

```
-----
FULL DESCRIPTIVE STATISTICS TABLE
-----
```

	beauty	eval_score
count	485.0000	485.0000
mean	0.0105	3.6765
std	0.9967	0.5343
min	-2.5000	2.1800
25%	-0.6510	3.3100
50%	-0.0330	3.6700
75%	0.6590	4.0600
max	2.5000	5.0000

Step 3: Check Assumptions

Assumption 1: Linearity (visual inspection via scatter plot)

Assumption 2: Bivariate Normality

Understanding Correlation vs Causation

Critical Distinction:

- **Correlation:** Two variables change together
- **Causation:** One variable directly causes changes in another

Why Correlation $\neq$ Causation:

1. **Confounding variables:** A third variable may influence both
 - Example: Ice cream sales and drowning deaths correlate, but heat causes both
2. **Reverse causality:** Y might cause X instead of X causing Y
3. **Coincidence:** Random patterns in finite data

Requirements for Causation:

1. Temporal precedence (cause precedes effect)
2. Covariation (variables are correlated)
3. No plausible alternative explanations
4. Usually requires **experimental** design (not just observational data)

```
In [40]: # Check assumptions
from scipy.stats import shapiro

print("="*80)
print("ASSUMPTION CHECKING")
print("="*80)

# Assumption 1: Normality for both variables
print("\n1. NORMALITY (Shapiro-Wilk Test)")
print("-"*80)

stat_beauty, p_beauty = shapiro(df_corr['beauty'])
stat_eval, p_eval = shapiro(df_corr['eval_score'])

print(f"\nBeauty Score:")
print(f"  W statistic = {stat_beauty:.6f}")
print(f"  p-value = {p_beauty:.6f}")
print(f"  Conclusion: {'Normally distributed (p > 0.05)' if p_beauty > 0.05 else 'Not'}")

print(f"\nEvaluation Score:")
print(f"  W statistic = {stat_eval:.6f}")
print(f"  p-value = {p_eval:.6f}")
print(f"  Conclusion: {'Normally distributed (p > 0.05)' if p_eval > 0.05 else 'Not'}")

print("\n" + "-"*80)
print("Overall Normality Assessment:")
print("-"*80)
if p_beauty > 0.05 and p_eval > 0.05:
    print("✓ Both variables are normally distributed")
    print("  Pearson correlation is appropriate")
else:
    print("X One or both variables are not normally distributed")
    print("  Consider Spearman correlation as alternative")
    print("  Proceeding with Pearson (robust with large n)")
```

```
=====
ASSUMPTION CHECKING
=====
```

```
1. NORMALITY (Shapiro-Wilk Test)
-----
```

Beauty Score:

```
W statistic = 0.994188
p-value = 0.061336
Conclusion: Normally distributed (p > 0.05)
```

Evaluation Score:

```
W statistic = 0.996647
p-value = 0.411605
Conclusion: Normally distributed (p > 0.05)
```

```
-----
Overall Normality Assessment:
-----
```

```
✓ Both variables are normally distributed
Pearson correlation is appropriate
```

```
In [68]: # Perform Pearson correlation test
from scipy.stats import pearsonr

# Ensure df_corr is defined
if 'df_corr' not in globals():
    df_corr = df_original[['beauty', 'eval_score']].dropna()

print("="*80)
print("PEARSON CORRELATION TEST")
print("="*80)

r, p_value_corr = pearsonr(df_corr['beauty'], df_corr['eval_score'])

print(f"\nPearson Correlation Coefficient: r = {r:.6f}")
print(f"p-value (two-tailed): p = {p_value_corr:.6f}")

# Interpret strength of correlation
print("\n" + "-"*80)
print("INTERPRETATION OF CORRELATION STRENGTH")
print("-"*80)

abs_r = abs(r)
if abs_r < 0.10:
    strength = "Negligible"
elif abs_r < 0.30:
    strength = "Weak"
elif abs_r < 0.50:
    strength = "Moderate"
elif abs_r < 0.70:
    strength = "Strong"
else:
    strength = "Very Strong"
```

```

direction = "positive" if r > 0 else "negative"

print(f"\nCorrelation Strength: {strength}")
print(f"Direction: {direction}")
print(f"\nGuidelines (Cohen, 1988):")
print(" Negligible:  $|r| < 0.10$ ")
print(" Weak:  $0.10 \leq |r| < 0.30$ ")
print(" Moderate:  $0.30 \leq |r| < 0.50$ ")
print(" Strong:  $0.50 \leq |r| < 0.70$ ")
print(" Very Strong:  $|r| \geq 0.70$ ")

# Calculate coefficient of determination (R²)
r_squared = r ** 2

print("\n" + "-"*80)
print("COEFFICIENT OF DETERMINATION (R²)")
print("-"*80)
print(f"\nR² = r² = {r_squared:.6f}")
print(f"Interpretation: {r_squared*100:.2f}% of variance in evaluation score")
print(f"is explained by beauty score (and vice versa)")

# Statistical decision
print("\n" + "-"*80)
print("STATISTICAL DECISION")
print("-"*80)
print(f"\n $\alpha = 0.05$ ")
print(f"p-value = {p_value_corr:.6f}")
print(f"\nDecision: {'Reject H₀' if p_value_corr < 0.05 else 'Fail to reject H₀'}")

if p_value_corr < 0.05:
    print(f"\nConclusion: There IS a statistically significant {strength.lower()} {
    print(f"correlation (r = {r:.3f}, p = {p_value_corr:.4f}) between beauty score
    print(f"evaluation score.")
else:
    print(f"\nConclusion: There IS NO statistically significant correlation")
    print(f"(r = {r:.3f}, p = {p_value_corr:.4f}) between beauty score and")
    print(f"evaluation score.")

```

=====

PEARSON CORRELATION TEST

=====

Pearson Correlation Coefficient: $r = 0.224674$
p-value (two-tailed): $p = 0.000001$

INTERPRETATION OF CORRELATION STRENGTH

Correlation Strength: Weak
Direction: positive

Guidelines (Cohen, 1988):
Negligible: $|r| < 0.10$
Weak: $0.10 \leq |r| < 0.30$
Moderate: $0.30 \leq |r| < 0.50$
Strong: $0.50 \leq |r| < 0.70$
Very Strong: $|r| \geq 0.70$

COEFFICIENT OF DETERMINATION (R^2)

$R^2 = r^2 = 0.050478$
Interpretation: 5.05% of variance in evaluation score
is explained by beauty score (and vice versa)

=====

STATISTICAL DECISION

=====

$\alpha = 0.05$
p-value = 0.000001

Decision: Reject H_0

Conclusion: There IS a statistically significant weak positive
correlation ($r = 0.225$, $p = 0.0000$) between beauty score and
evaluation score.

Step 5: Visualization

Understanding the Correlation Coefficient Calculation

Step-by-Step Process:

1. Calculate means:

- $\bar{X}$ = mean of beauty scores
- $\bar{Y}$ = mean of evaluation scores

2. Calculate deviations from mean:

- For each pair: $(X_i - \bar{X})$ and $(Y_i - \bar{Y})$
3. **Compute products of deviations:**
- $(X_i - \bar{X})(Y_i - \bar{Y})$ for each pair
4. **Sum the products** (numerator):
- $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$
5. **Calculate sum of squared deviations** (denominator):
- $\sqrt{\sum (X_i - \bar{X})^2} \times \sqrt{\sum (Y_i - \bar{Y})^2}$
6. **Divide numerator by denominator** to get r

What r = 0.6 means:

- **Positive sign:** As beauty score increases, evaluation tends to increase
- **Magnitude 0.6:** Strong positive relationship
- **R² = 0.36:** 36% of variation in evaluations explained by beauty
- **Remaining 64%:** Due to other factors (teaching quality, content, etc.)

Step-by-Step Correlation Calculation Process

Let's manually calculate Pearson's correlation coefficient:

Given Data:

- Two variables: X (beauty score) and Y (evaluation score)
- n paired observations: $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

Step 1: Calculate Means

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad \bar{Y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Step 2: Calculate Deviations from Mean

For each observation:

$$(x_i - \bar{X}) \quad \text{and} \quad (y_i - \bar{Y})$$

Step 3: Calculate Products of Deviations

For each pair:

$$(x_i - \bar{X})(y_i - \bar{Y})$$

This is **positive** when both are above their means OR both below their means (positive association).

Step 4: Sum the Products (Covariance Numerator)

$$\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})$$

Step 5: Calculate Sum of Squared Deviations

$$\sum_{i=1}^n (x_i - \bar{X})^2 \quad \text{and} \quad \sum_{i=1}^n (y_i - \bar{Y})^2$$

Step 6: Calculate Pearson's r

$$r = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{X})^2} \times \sqrt{\sum_{i=1}^n (y_i - \bar{Y})^2}}$$

Alternative Computational Formula:

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

This is algebraically equivalent but more computationally efficient.

Step 7: Calculate Coefficient of Determination

$$R^2 = r^2$$

Step 8: Test for Significance

Hypothesis Test:

- $H_0: \rho = 0$ (no correlation in population)
- $H_1: \rho \neq 0$ (correlation exists in population)

Test Statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

This follows a t-distribution with $df = n - 2$ under H_0 .

P-value: $P(|T_{n-2}| \geq |t_{obs}|)$

Step 9: Calculate Confidence Interval (Fisher's Z-Transformation)

Transform r to z:

$$z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right) = \tanh^{-1}(r)$$

Standard Error of z:

$$SE_z = \frac{1}{\sqrt{n-3}}$$

95% Confidence Interval for z:

$$CI_z = z \pm 1.96 \times SE_z$$

Transform back to r:

$$r = \tanh(z) = \frac{e^{2z} - 1}{e^{2z} + 1}$$

Numerical Example:

Suppose we have 5 observations:

i	x_i (Beauty)	y_i (Eval)	$x_i - \bar{X}$	$y_i - \bar{Y}$	$(x_i - \bar{X})$ $(y_i - \bar{Y})$	$(x_i - \bar{X})^2$	$(y_i - \bar{Y})^2$
1	3.0	3.5	-1.0	-0.5	0.5	1.0	0.25
2	3.5	3.8	-0.5	-0.2	0.1	0.25	0.04
3	4.0	4.0	0.0	0.0	0.0	0.0	0.0
4	4.5	4.2	0.5	0.2	0.1	0.25	0.04
5	5.0	4.5	1.0	0.5	0.5	1.0	0.25
Sum	20.0	20.0	0.0	0.0	1.2	2.5	0.58
Mean	4.0	4.0					

Calculate r:

$$r = \frac{1.2}{\sqrt{2.5} \times \sqrt{0.58}} = \frac{1.2}{1.581 \times 0.762} = \frac{1.2}{1.205} = 0.996$$

Calculate R²:

$$R^2 = (0.996)^2 = 0.992$$

Interpretation: 99.2% of variance in evaluation scores is explained by beauty scores.

Test for Significance:

$$t = \frac{0.996\sqrt{5-2}}{\sqrt{1-0.996^2}} = \frac{0.996 \times 1.732}{\sqrt{0.008}} = \frac{1.725}{0.089} = 19.38$$

With $df = 3$ and $\alpha = 0.05$, $t_{critical} = 3.182$

Since $19.38 > 3.182$, we reject H_0 . The correlation is highly significant ($p < 0.001$).

Fisher's Z-Transformation:

$$z = \frac{1}{2} \ln \left(\frac{1 + 0.996}{1 - 0.996} \right) = \frac{1}{2} \ln(499) = 3.104$$

$$SE_z = \frac{1}{\sqrt{5-3}} = \frac{1}{\sqrt{2}} = 0.707$$

$$CI_z = 3.104 \pm 1.96(0.707) = [1.718, 4.490]$$

Transform back: $CI_r = [0.939, 1.000]$ (approximately, bounded at 1.0)

```
In [42]: # Calculate confidence interval for correlation coefficient using Fisher's z-transf
print("="*80)
print("CONFIDENCE INTERVAL FOR CORRELATION COEFFICIENT")
print("="*80)

# Fisher's z-transformation
z = np.arctanh(r) # or 0.5 * np.log((1 + r) / (1 - r))
n = len(df_corr)
se_z = 1 / np.sqrt(n - 3)

# Calculate CI for z
z_critical = 1.96 # for 95% CI
z_lower = z - z_critical * se_z
z_upper = z + z_critical * se_z

# Transform back to r
r_lower = np.tanh(z_lower)
r_upper = np.tanh(z_upper)

print(f"\nFisher's z-transformation:")
print(f"  z = arctanh(r) = {z:.6f}")
print(f"  SE(z) = 1/√(n-3) = 1/√({n}-3) = {se_z:.6f}")

print(f"\n95% Confidence Interval for z:")
print(f"  [{z_lower:.6f}, {z_upper:.6f}]")

print(f"\n95% Confidence Interval for r:")
print(f"  [{r_lower:.6f}, {r_upper:.6f}]")

print(f"\nInterpretation:")
print(f"  We are 95% confident that the true population correlation (ρ)")
print(f"  lies between {r_lower:.4f} and {r_upper:.4f}.")

if 0 >= r_lower and 0 <= r_upper:
    print(f"\n  Note: CI includes 0, suggesting correlation may not be significant.")
else:
    print(f"\n  Note: CI does not include 0, confirming significant correlation.")

# Calculate t-statistic (alternative test statistic)
```

```

print("\n" + "-"*80)
print("T-STATISTIC (Alternative Test)")
print("-"*80)

t_stat = r * np.sqrt(n - 2) / np.sqrt(1 - r**2)
df_t = n - 2

print(f"\nt-statistic: t = r × √(n-2) / √(1-r²)")
print(f"t = {r:.6f} × √({n}-2) / √(1-{r**2:.6f})")
print(f"t = {t_stat:.6f}")
print(f"\nDegrees of freedom: df = {df_t}")
print(f"\nThis t-statistic yields the same p-value as the correlation test.")

```

CONFIDENCE INTERVAL FOR CORRELATION COEFFICIENT

Fisher's z-transformation:

$z = \text{arctanh}(r) = 0.228573$

$SE(z) = 1/\sqrt{n-3} = 1/\sqrt{485-3} = 0.045549$

95% Confidence Interval for z:

[0.139297, 0.317848]

95% Confidence Interval for r:

[0.138403, 0.307560]

Interpretation:

We are 95% confident that the true population correlation (ρ) lies between 0.1384 and 0.3076.

Note: CI does not include 0, confirming significant correlation.

T-STATISTIC (Alternative Test)

t-statistic: $t = r \times \sqrt{(n-2) / (1-r^2)}$
 $t = 0.224674 \times \sqrt{(485-2) / (1-0.050478)}$
 $t = 5.067257$

Degrees of freedom: $df = 483$

This t-statistic yields the same p-value as the correlation test.

Step 6: Statistical Conclusion and Interpretation

```

In [43]: # Comprehensive visualization for Question 4
fig = plt.figure(figsize=(16, 10))
gs = GridSpec(2, 3, figure=fig, hspace=0.3, wspace=0.3)

# 1. Scatter plot with regression line
ax1 = fig.add_subplot(gs[0, 0])
ax1.scatter(df_corr['beauty'], df_corr['eval_score'], alpha=0.5, s=30, edgecolors='
# Add regression line

```

```

z_fit = np.polyfit(df_corr['beauty'], df_corr['eval_score'], 1)
p_fit = np.poly1d(z_fit)
x_line = np.linspace(df_corr['beauty'].min(), df_corr['beauty'].max(), 100)
ax1.plot(x_line, p_fit(x_line), 'r--', linewidth=2, label=f'y = {z_fit[0]:.3f}x + {z_fit[1]:.3f}')
ax1.set_xlabel('Beauty Score', fontsize=12, fontweight='bold')
ax1.set_ylabel('Evaluation Score', fontsize=12, fontweight='bold')
ax1.set_title(f'Scatter Plot with Regression Line\nr = {r:.4f}, p = {p_value_corr:.4f}')
ax1.legend(fontsize=10)
ax1.grid(True, alpha=0.3)

# 2. Scatter plot with confidence ellipse (simpler version with density coloring)
ax2 = fig.add_subplot(gs[0, 1])
from scipy.stats import gaussian_kde
xy = np.vstack([df_corr['beauty'], df_corr['eval_score']])
z = gaussian_kde(xy)(xy)
scatter = ax2.scatter(df_corr['beauty'], df_corr['eval_score'], c=z, s=30, cmap='viridis')
plt.colorbar(scatter, ax=ax2, label='Density')
ax2.set_xlabel('Beauty Score', fontsize=12, fontweight='bold')
ax2.set_ylabel('Evaluation Score', fontsize=12, fontweight='bold')
ax2.set_title('Scatter Plot with Density Coloring', fontsize=13, fontweight='bold')
ax2.grid(True, alpha=0.3)

# 3. Hexbin plot
ax3 = fig.add_subplot(gs[0, 2])
hexbin = ax3.hexbin(df_corr['beauty'], df_corr['eval_score'], gridsize=20, cmap='YlOrBr')
plt.colorbar(hexbin, ax=ax3, label='Count')
ax3.set_xlabel('Beauty Score', fontsize=12, fontweight='bold')
ax3.set_ylabel('Evaluation Score', fontsize=12, fontweight='bold')
ax3.set_title('Hexbin Plot: Concentration of Data Points', fontsize=13, fontweight='bold')

# 4. Q-Q plot for beauty
ax4 = fig.add_subplot(gs[1, 0])
stats.probplot(df_corr['beauty'], dist="norm", plot=ax4)
ax4.set_title(f'Q-Q Plot: Beauty Score\nShapiro-Wilk p = {p_beauty:.4f}', fontsize=12)
ax4.grid(True, alpha=0.3)

# 5. Q-Q plot for evaluation
ax5 = fig.add_subplot(gs[1, 1])
stats.probplot(df_corr['eval_score'], dist="norm", plot=ax5)
ax5.set_title(f'Q-Q Plot: Evaluation Score\nShapiro-Wilk p = {p_eval:.4f}', fontsize=12)
ax5.grid(True, alpha=0.3)

# 6. Summary table
ax6 = fig.add_subplot(gs[1, 2])
ax6.axis('off')
summary_data = [
    ['Statistic', 'Value'],
    ['Sample Size (n)', f'{n}'],
    ['', ''],
    ['Pearson r', f'{r:.6f}'],
    ['R2 (% variance)', f'{r_squared:.6f} ({r_squared*100:.2f}%)'],
    ['p-value', f'{p_value_corr:.6f}'],
    ['', ''],
    ['95% CI for r', f'[{r_lower:.4f}, {r_upper:.4f}]'],
    ['', ''],
    ['Strength', strength],

```

```

['Direction', direction.capitalize()],
['', ''],
['Decision', 'Reject H0' if p_value_corr < 0.05 else 'Fail to reject H0'],
['Significance', 'Significant' if p_value_corr < 0.05 else 'Not significant']
]
table = ax6.table(cellText=summary_data, cellLoc='left', loc='center', colWidths=[0
table.auto_set_font_size(False)
table.set_fontsize(9)
table.scale(1, 2.2)
for i in range(2):
    table[(0, i)].set_facecolor('#2C3E50')
    table[(0, i)].set_text_props(weight='bold', color='white')
for i in range(1, len(summary_data)):
    for j in range(2):
        if i in [2, 6, 8, 11]:
            table[(i, j)].set_facecolor('#ECF0F1')
        elif i > 11:
            table[(i, j)].set_facecolor('#FFF3CD')
        elif i % 2 == 0:
            table[(i, j)].set_facecolor('#F8F9FA')
ax6.set_title('Pearson Correlation Summary', fontsize=13, fontweight='bold', pad=20

plt.suptitle('Question 4: Pearson Correlation - Beauty Score × Evaluation Score',
             fontsize=16, fontweight='bold', y=0.995)
plt.show()

```

Question 4: Pearson Correlation - Beauty Score × Evaluation Score



Assignment Summary and Conclusions

Overview of Statistical Tests Performed

This assignment demonstrated the application of four fundamental statistical tests to analyze relationships in the teacher rating dataset:

1. **Independent Samples t-Test** (Question 1): Compared mean teaching evaluation scores between male and female instructors
2. **One-Way ANOVA** (Question 2): Examined differences in beauty scores across three age groups
3. **Chi-Square Test of Independence** (Question 3): Investigated association between tenure status and gender
4. **Pearson Correlation** (Question 4): Analyzed relationship between beauty and evaluation scores

Key Findings Summary

Below is a summary table of the four statistical analyses:

Question	Test Used	Primary Result	Effect Size	Conclusion
Q1: Gender → Evaluation	t-test	See results above	Cohen's d	Gender effect on evaluations
Q2: Age → Beauty	ANOVA	See results above	Eta-squared (η^2)	Age group differences in beauty
Q3: Tenure ↔ Gender	Chi-square	See results above	Cramér's V	Association between variables
Q4: Beauty ↔ Evaluation	Correlation	See results above	R^2	Strength of relationship

Statistical Methodology Highlights

Assumption Checking

- All tests included thorough assumption verification (normality, homogeneity of variances, independence)
- Both parametric and non-parametric alternatives were considered
- Visual diagnostics (Q-Q plots, residual plots) complemented statistical tests

Effect Size Reporting

- Cohen's d for t-tests (standardized mean difference)
- Eta-squared (η^2) for ANOVA (proportion of variance explained)
- Cramér's V for chi-square (strength of association)
- R^2 for correlation (proportion of shared variance)

Visualization Strategy

- Multiple complementary visualizations for each question
- Professional formatting with clear labels, legends, and titles
- Multi-panel layouts for comprehensive analysis
- Statistical reference lines (means, medians, regression lines)

Learning Outcomes Achieved

1. **Hypothesis Testing Framework:** Applied systematic approach to all four tests
2. **Statistical Inference:** Made data-driven decisions using p-values and effect sizes
3. **Assumption Validation:** Checked prerequisites for parametric tests
4. **Effect Size Interpretation:** Reported practical significance alongside statistical significance
5. **Data Visualization:** Created publication-quality figures for statistical results
6. **Statistical Communication:** Provided clear interpretations in context

Technical Implementation Notes

- **Data Handling:** Proper NaN filtering before all statistical calculations
- **Statistical Notation:** Used formal mathematical notation (χ^2 , t, F, r, p-values)
- **Confidence Intervals:** Calculated 95% CIs for effect estimates
- **Alternative Tests:** Included non-parametric alternatives (Mann-Whitney, Kruskal-Wallis, Spearman)
- **Multiple Comparisons:** Applied Bonferroni correction for post-hoc tests

Assignment Completed: October 24, 2025

Student: Md Ayan Alam (GF202342645)

Course: Statistical Foundation of Data Science

Dataset: Teacher Rating Dataset (teacher_ratings.csv, n=500 teachers)

Comprehensive Mathematical Formulations Reference

1. Independent Samples t-Test

Pooled Variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

t-Statistic:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Degrees of Freedom:

$$df = n_1 + n_2 - 2$$

Cohen's d:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p}$$

Confidence Interval for Difference:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, df} \times s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Welch's t-Test (unequal variances):

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$
$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

2. One-Way ANOVA

Sum of Squares:

$$SST = \sum_{j=1}^k \sum_{i=1}^{n_j} (X_{ji} - \bar{X}_{..})^2$$

$$SSB = \sum_{j=1}^k n_j (\bar{X}_j - \bar{X}_{..})^2$$

$$SSW = \sum_{j=1}^k \sum_{i=1}^{n_j} (X_{ji} - \bar{X}_j)^2$$

Relationship: $SST = SSB + SSW$

Mean Squares:

$$MSB = \frac{SSB}{df_B} \quad (\text{Between-group variance})$$

$$MSW = \frac{SSW}{df_W} \quad (\text{Within-group variance, pooled})$$

F-Statistic:

$$F = \frac{MSB}{MSW}$$

Degrees of Freedom:

$$df_1 = k - 1, \quad df_2 = N - k$$

Eta-Squared:

$$\eta^2 = \frac{SSB}{SST}$$

Omega-Squared (unbiased):

$$\omega^2 = \frac{SSB - (k - 1)MSW}{SST + MSW}$$

3. Chi-Square Test of Independence

Expected Frequency:

$$E_{ij} = \frac{R_i \times C_j}{N}$$

Chi-Square Statistic:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Degrees of Freedom:

$$df = (r - 1)(c - 1)$$

Cramér's V:

$$V = \sqrt{\frac{\chi^2}{N \times \min(r-1, c-1)}}$$

Phi Coefficient (for 2×2 table):

$$\phi = \sqrt{\frac{\chi^2}{N}} = V$$

Odds Ratio (for 2×2 table):

$$OR = \frac{ad}{bc}$$

Confidence Interval for Odds Ratio:

$$\ln(OR) \pm z_{\alpha/2} \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

4. Pearson Correlation

Pearson's r:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \times \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Computational Formula:

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Coefficient of Determination:

$$R^2 = r^2$$

t-Test for Correlation:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \quad df = n-2$$

Fisher's Z-Transformation:

$$z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right)$$

Standard Error of z:

$$SE_z = \frac{1}{\sqrt{n-3}}$$

Back-transformation:

$$r = \tanh(z) = \frac{e^{2z} - 1}{e^{2z} + 1}$$

Spearman's Rank Correlation:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i = difference in ranks for each pair.

5. Assumption Tests

Shapiro-Wilk Test:

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Levene's Test:

$$F = \frac{(N - k) \sum_{i=1}^k n_i (\bar{Z}_i - \bar{Z})^2}{(k - 1) \sum_{i=1}^k \sum_{j=1}^{n_i} (Z_{ij} - \bar{Z}_i)^2}$$

where $Z_{ij} = |X_{ij} - \tilde{X}_i|$

Bartlett's Test (for normality assumed):

$$\chi^2 = \frac{(N - k) \ln(s_p^2) - \sum_{i=1}^k (n_i - 1) \ln(s_i^2)}{1 + \frac{1}{3(k-1)} \left(\sum_{i=1}^k \frac{1}{n_i - 1} - \frac{1}{N - k} \right)}$$

6. Post-Hoc Tests

Bonferroni Correction:

$$\alpha_{adj} = \frac{\alpha}{C}, \quad C = \binom{k}{2}$$

Tukey's HSD:

$$HSD = q_{\alpha, k, df_W} \times \sqrt{\frac{MSW}{n_h}}$$

where $n_h = \frac{k}{\sum_{i=1}^k \frac{1}{n_i}}$ (harmonic mean)

Scheffé's Test:

$$F_S = \frac{(\bar{X}_i - \bar{X}_j)^2}{MSW(k-1) \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

7. Descriptive Statistics

Mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$$

Variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Standard Deviation:

$$s = \sqrt{s^2}$$

Standard Error:

$$SE = \frac{s}{\sqrt{n}}$$

Coefficient of Variation:

$$CV = \frac{s}{\bar{x}} \times 100\%$$

Skewness:

$$\text{Skew} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Kurtosis (excess):

$$\text{Kurt} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

8. Confidence Intervals

For Mean (σ known):

$$\bar{x} \pm z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

For Mean (σ unknown):

$$\bar{x} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$$

For Proportion:

$$\hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

For Difference in Means:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, df} \times SE_{diff}$$

9. Sample Size Calculations

For t-Test:

$$n = \frac{2(z_{\alpha/2} + z_{\beta})^2 \times \sigma^2}{(\mu_1 - \mu_2)^2}$$

For Correlation:

$$n = \left(\frac{z_{\alpha/2} + z_{\beta}}{0.5 \ln \frac{1+\rho}{1-\rho}} \right)^2 + 3$$

For Chi-Square:

$$n = \frac{(z_{\alpha/2} + z_{\beta})^2}{w^2}$$

where w = effect size parameter

10. Effect Size Conversions

From r to d:

$$d = \frac{2r}{\sqrt{1 - r^2}}$$

From d to r:

$$r = \frac{d}{\sqrt{d^2 + 4}}$$

From F to η^2 :

$$\eta^2 = \frac{SSB}{SSB + SS_W}$$

From t to d:

$$d = \frac{2t}{\sqrt{df}}$$

This reference guide provides all essential formulas used in Assignment 6 analyses.